PEARLAND FIRE DEPARTMENT

Engine Operations Manual

"Vincit Semper Aquam"

It is the philosophy of the Pearland Fire Department Engine Company Operations Manual to establish a set standard of philosophies to apply consistently throughout our organization. This manual provides guidance allowing PFD to operate with aggressive yet calculated tactics in our fire attack operations. To establish the "Pearland Way" and setting clear expectations and understanding for "yet calculated" is that we expect all members, and we will hold each other to the standard of aggressively performing all tasks, duties, and assignments to the full extent possible within our physical abilities, protective equipment, and training. This allows us to achieve the highest level of success within our goals of preservation of life, scene stabilization, and property conservation. Utilizing the philosophies within this manual, PFD Engine Companies will function in a professional mindset and routinely display their technical excellence with the goal of mitigating any obstacle that stands in our way. The Engine Companies' most important duty, primary task, and greatest amount of training time should be dedicated to the task of expertly and proficiently pulling and advancing the correct hose line to the correct location and to "Get water on the fire".

It is our desire to advance PFD collectively in an upward trajectory of professionalism and technical prowess that others want to emulate. Our progressive display of strategies and tactics will allow us to perform more efficiently and in a safer manner. This aggressive, yet professionally aggressive mentality is what we refer to as "The Pearland Way". Be proud to be recognized as trendsetters and part of a unique group of talented firefighters, nozzle men and women, that others can only hope to come close to equaling.

J. Taylor
Chief of Department

By Orders of:

J. Taylor - Chief of the Department

C. Birt - Assistant Chief of Operations

J. Johnson - Assistant Chief of Administration

Effective This Day: 31st July 2024

This 2nd Edition of the PFD Engine Company Operations Manual has been reviewed and approved by Command Staff effective this day: 24 March 2025

C. Birt - Assistant Chief of Operations

Table of Contents

Section I: History of the PFD Engine Company

The Pearland Fire Department comes from modest roots. This is common in suburban areas surrounding the metropolis of Southeast Texas. Like many of those departments, Pearland was forced to undertake massive growth quickly. Since the inception of the first all-volunteer Engine in 1946 to the 24/7 engine companies of today, a lot has changed. The correlation between department growth and the city's population goes hand in hand. From 1946 until 2007, the Pearland Fire Department was operated strictly by volunteers. In 2007 the first full-time staffing was placed into operation with two apparatus, one being an Engine Company located at the corner of Orange and Old Alvin. Soon it was known that the dramatic development of Pearland would require additional staffing. As the number of firehouses and firefighters increased, the department merged with the already existing EMS department in 2014. Throughout the years, the culture within the department has been to provide aggressive firefighting that places the public's safety paramount. As with any department growth, so comes department experience. Because of the rapid intake of firefighters, the Pearland Fire Department has a plethora of knowledge and expertise across all ranks. Taking this coupled with science-backed tactics, Pearland Fire Department has assembled an industry-leading Engine Company culture. The department intends that all members assigned to an Engine have the best equipment, best training, and are entrusted with the best abilities. Engine Company operations are the fundamental cornerstone of American firefighting. All other fireground work is built around the Engine Company achieving its goal. To get water on the fire and put the fire out. When the fire is suppressed, all other actions on the fireground become safer for all involved, especially our residents. The number one priority of all Pearland Fire Department Engine Companies is to put the fire out. It is understood and accepted that the fire goes as the first line goes. Engine Companies are expected and entrusted to be valiant and well-trained in stretching lines, proper pumping and to be the ones who provide the safety aspect of flowing water. Engine company members must immerse themselves into the engine culture and understand that everything on the fireground revolves around putting the fire out.

Section II: Definitions

Attack Team – A team under the command of a Company Officer equipped with a handline for the purposes of fire attack.

Attack line- a hose line used to apply water directly onto a fire and operated by a sufficient number of personnel so that it can be maneuvered effectively and safely.

Backup line- a hose line that is of equal or greater size/flow to the initial attack line that is deployed for the sole purpose of providing protection to the initial attack line. This line will be positioned to protect the egress of the initial attack line or to protect the initial attack line from adverse fire conditions.

Booster Tank- The onboard water tank on the apparatus. 750 gallons for Engines.

Cooling – The desired effect of reduction of actual and perceived temperature in the involved structure as a result of a properly flowing handline

Engine – A fire apparatus equipped with a pump, booster tank, and hose for the purposes of fire attack. A fully equipped engine includes all equipment listed in the equipment list in section one of this manual.

Engine Company – The engine and its personnel. A four-person company consisting of a company Officer, driver/operator, and two firefighters.

Initial Attack Line - The first hose stream placed in service by a company at the scene of a fire to protect lives or to prevent further extension of fire while additional lines are laid and placed in position.

Lift – The final indicator of an effective stream. The visible thermal barrier will lift due to heated gases' cooling and subsequent contraction. This is a late indicator and is dependent on ventilation conditions.

Return – The return of water from a hose stream after hitting the ceiling and returning to the ground. This is an indicator of a properly flowing stream. The stream passing through the superheated gasses, broken against a solid surface, and passing back through has the most significant cooling effect.

Secondary Line – An additional attack line, typically pulled by the second-due engine company, this line is used to attack the fire in a different geographic location (ex: A-Side vs D-Side) than the initial attack line. Note: This is different from a Backup Line which is used to protect the initial attack team's path of egress.

Stacking tanks – The process of joining 2 engine companies booster tanks for rapid augmentation of water supply. When stacking tanks, 2 ½" hose should be utilized to reduce the water lost in the hose.

Section III: Engine Company Equipment

The following equipment will be the minimum equipment found on all PFD Engine companies. All PFD Pumpers and apparatus will always be set up for offensive operations. All deck guns, monitors and ladder pipes will be equipped with smooth bore nozzles for immediate needs. Changes to other nozzles can be done if needed for special operations.

- Adapters
  - 1.5-inch double male
  - 1.5-inch double female
  - 1.5-inch female to 1-inch male.
  - 2.5-inch double male x2
  - 2.5-inch double female x2
  - 2.5-inch female to 1-inch male. x 2
  - 2.5 inch to 1.5-inch reducer x2
  - 2.5 inch to garden hose reducer
  - Storz to 4-inch female
  - Storz to 4.5-inch female adapter
  - Storz to 2.5-inch male.
- Appliances
  - Gated wye
  - 2.5-inch valve
  - Eductor
- Hand Tools
  - Push broom x 2
  - Scoop shovel
  - Flathead axe x 2
  - Halligan bar x 2
  - Trash hook
  - 6’ NY hook x 2
  - 8’ NY hook
  - 10’ NY Hook
  - Bolt cutters 18”
  - Officer tool
- Hydrant bag (Klein Tool canvas bag)
  - 4” Storz Adapter
  - Gated valve for 2.5” hydrant connection.
  - LDH Spanners x2
  - SDH Spanners x2
  - Dead blow mallet
  - Hydrant Wrench
- Dead blow mallet
- Decon line with nozzle
- Elevator keys
- Hot stick
- K tool
- Toolkit
- Ground monitor – 1 3/8 SS tips
- Pipe wrench
- Spanners / Wrenches (additional to hydrant bag)
  - Storz spanner x 4
  - Small spanner x 4
  - Hydrant wrench x 2
- 5” hose clamp
- Traffic cones x 5
- Disposable tarps x 2
- Cooler
- Caution tape
- Ladders
  - 24’ Extension

  - 14’ roof
  - 10’ attic
- Glass master
- Floor dry (5 Gallon Bucket)
- Water can w / strap x 2
- CO_2 extinguisher
- Dry chem extinguisher
- Step chocks x 2
- Big Easy
- Life jacket / personal flotation device x 4
- Water rescue throw bag x2
- SCBA bottle x 6
- Smoke detector backpack
- Quick struts x 2 (with j hooks, chains and accessories)
- Hose
  - 700’ 5” (100’ lengths)
  - 100’ 5” (50’ lengths)
  - 50’ 5” (25’ lengths)
  - 600’ 2 ½” w/ nozzle (1 3/16 integrated tip with 1 1/8 tip)
  - 300’ 2 ½” with gated wye and 100’ of 1 ¾”
  - 300’ 2 ½” with nozzle and 100’ 1 ¾”
  - 50’ 2 ½” in saddle tray
  - 100’ 1 ¾” with 7/8” nozzle (bumper)
  - 100’ 1” forestry hose (bumper)
  - Gate-Wye /w 1.5-inch female to 1-inch male adapter
  - 150’ of 2 ½" for Denver Loads located in speedlay chutes or designated area.
- Nozzles (for each line and a spare of each)
  - 2 ½ nozzles
    - Integrated 1 3/16” with 1 1/8” tip
  - 1 ¾ nozzles
    - 7/8 tip
  - Forestry nozzle
  - 160 GPM fixed gallonage fog tip (in DO compartment for use when needed)
- Hose straps
  - 3 straps per bundle
- Chain saw
- Cutoff Saw
- Battery powered tools
  - Drill / driver
  - Impact wrench
  - Reciprocating saw
- Battery Powered Combi Tool
  - Current tools will remain until replacement. Cutters and spreaders, no ram.
- Electric PPV Fan
- F500 Kit
  - TKO nozzle (1”female to 1 ½ male adapter)
  - 100’ forestry hose
  - F500 Bottle
  - 2 ½ to 1” reducer
- Standpipe Kit
  - 2.5” x 2.5” Gate Valve
  - 2.5” x 2.5” In-Line Analog Pressure Gauge
  - 2.5” x 2.5” 60-degree elbow with drain valve.
  - 2.5” Lightweight Cap.
  - Fast Wrench
  - Wedges x5
- Attack Hose Pack

        - 50’ of 2.5” TRU-ID hose in a Denver Load.
        - Elkhart 1 3/16” Shutoff with 1 1/8” tip
    - Supply Hose Pack x2
        - 50’ of 2.5” TRU-ID hose in a Denver Load.
    - Rope Jug
        - 100’ of 3/8” Utility Rope
        - Large Carabiner x2
- Report Writing Toughbook/Rear MDT
- Box lights x 4
- Gas Meter w/ Charger
- TIC x2
- Airpacks x 4
- RIT Pack
- EMS equipment
    - Medical Bag
    - Monitor / AED
    -
    - C-Collar bag
    - Active Shooter bag
    - Video Laryngoscope
    - Handtevy Bag
    - Oxygen Cylinder
    - Ultrasound
- Radio Straps x4
- Radios w / hand mic x4
- Traffic Vests x5
- Ballistic Vests x4
- 911 Knox Key
- Knox Lock tool (Dogbone key)
- Headsets x6
- Insurance Card
- Binoculars
- Map Books
- ERG
- Portable lighting x2
- Soft Stretcher
- PR Supplies
- Deck gun tips
    - Solid (1 3/8”, 1 ½”, 1 ¾”, 2”)
    - Stream Straightener
- Accountability Passports
- Garage Door opener
- Chargers for rechargeable equipment
    - Gas meter
    - TIC
- Trash Bags
- Makita Charger
    - Spare Batteries x 2
- Turbo Flares
- Spare Chainsaw Chain x 2
- Bar Oil
- Tru-Fuel
- EV-Blanket

Note: As new equipment is purchased it will phase out old equipment in accordance with the current standard for the make and model of each type of equipment. Quantities may vary based on the spec of apparatus. For instance, the quantities for nozzles will vary based on the number of attack lines the engine is equipped with. Each attack line will be equipped with the appropriate nozzle and one spare 2 ½" and one spare 1 ¾" nozzle will be on hand in the compartment. **Any handline nozzle that has an integrated tip in the shutoff will not have another tip of equal or larger diameter or flow attached.**

Note: Some apparatus will maintain additional equipment based on additional duties assigned. For instance, E8 will maintain additional technical rescue equipment as it serves as a support unit for TW8. o No changes to the configuration of the hose beds or other equipment listed will be made to accommodate equipment not listed without the express permission of the Assistant Chief of Operations.

Section IV: Hose Loads

All hose loads on PFD Engines will be constructed of two distinct and identifiable components. The Bundle and the supply load. This will apply to all attack lines regardless of the location of the load. Some slight modifications will be needed for each apparatus, but the overall concept of operations will be the same. **The use of rear hose beds in deployment will be the only approved method of deploying lines to structure fires, this allows the most amount of flexibility and function.**

Simplicity is key to any operation. The hose loads and finishes are designed to be utilized for almost all possible scenarios where a line is to be stretched. A Nozzle Firefighter will perform the same stretch on every fire, the only variable being the distance travelled.

The occupancies below make up the highest percentage of occupancies encountered. The attack package offers a solution to all these occupancies:

- Short offset residential < 2000 square ft. 30’ – 50’ from the street.
- Long offset residential > 100’ from street
- Strip malls
- 2-3 story apartments with exterior or interior stairwells and hallways
- Big Box / Warehouse
- Mid – High rise and standpipe equipped buildings. Residential and commercial.

Bundle

The Bundle on all 1 ¾ handlines will be constructed the same regardless of their location on the apparatus. The design is to give the Nozzle Firefighter 100 feet of working length of hose. The 2.5" hose after the 100-foot bundle will be considered the supply load. The Bundle will be constructed in the following manner regardless of placement on the engine. **Note:** All hose loads <u>**will be**</u> constructed from tightly rolled hose. Failure to expel all water and air from the hose will result in poor-performing loads.

<u>Equipment Needed</u>

Below is the equipment needed to build the Bundle (Figure 1).

- 6' NY Hook
- (2) 50' sections of 1 ¾" Tru-ID.
- Rubber Mallet
- Pull Strap
- Three Bundle Straps

Figure 1

1. Place the 6’ NY Hook on the ground and place your coupling at the apex of the hook. (Figure 2). Run the hose down and make your first fold in line with the end of the hook.

Figure 2

2. After making your first fold in line with the bottom of the NY Hook, run the hose back on top of itself and make your next fold just before the coupling. (Figure 3). Continue making these same folds in Step 1 & 2 until all 50’ is folded in one stack. All folds should be equal with the folds below it. Any folds that don’t lay flat should either be manipulated by hand or hit with a rubber mallet until they lay flush so that all folds lay flat.
3. Once all 50’ of the first section of hose has been stacked on top of itself. The second 50’ roll should be connected to the first section. Once connected, continued to run the hose on top of the first section and make one last fold on top of the first stack.
4. Starting from the last fold on the first stack, beginning transition the hose down the side of the load until the hose begins to lay flat creating the base for the second stack. (Figure 4 & 5).

Figure 3

Figure 4

Figure 5

5. The first fold of the second stack should be parallel to the folds on the first stack. Continue making folds the same way as the first stack until the second 50’ roll is stacked on top of itself. If done properly, the 50’ coupling should lay in the middle of the first stack and the final coupling with the nozzle should be laying close to the middle.

6. As with previous iterations of the bundle load, the bundle will be secured together with three bundle straps with the addition of a pull strap. Place all three straps on the ground in the same orientation. Take the pull strap and thread the loops of the pull strap onto the middle strap and the strap that will be on the opposite side of the bundle from the female coupling. (Figure 6 & 7).

Figure 6

Figure 7

7. Once the straps have been layed out in the above fashion they can be slid underneath the hose load or the hose load built ontop of the straps before starting Step 1. Secure the bundle straps as tightly as possible, utilizing a rubber mallet if necessary. The straps

should be oriented in the same direction and on the top of the bundle. The bundle should be compact and tight on all sides. (Figure 8)

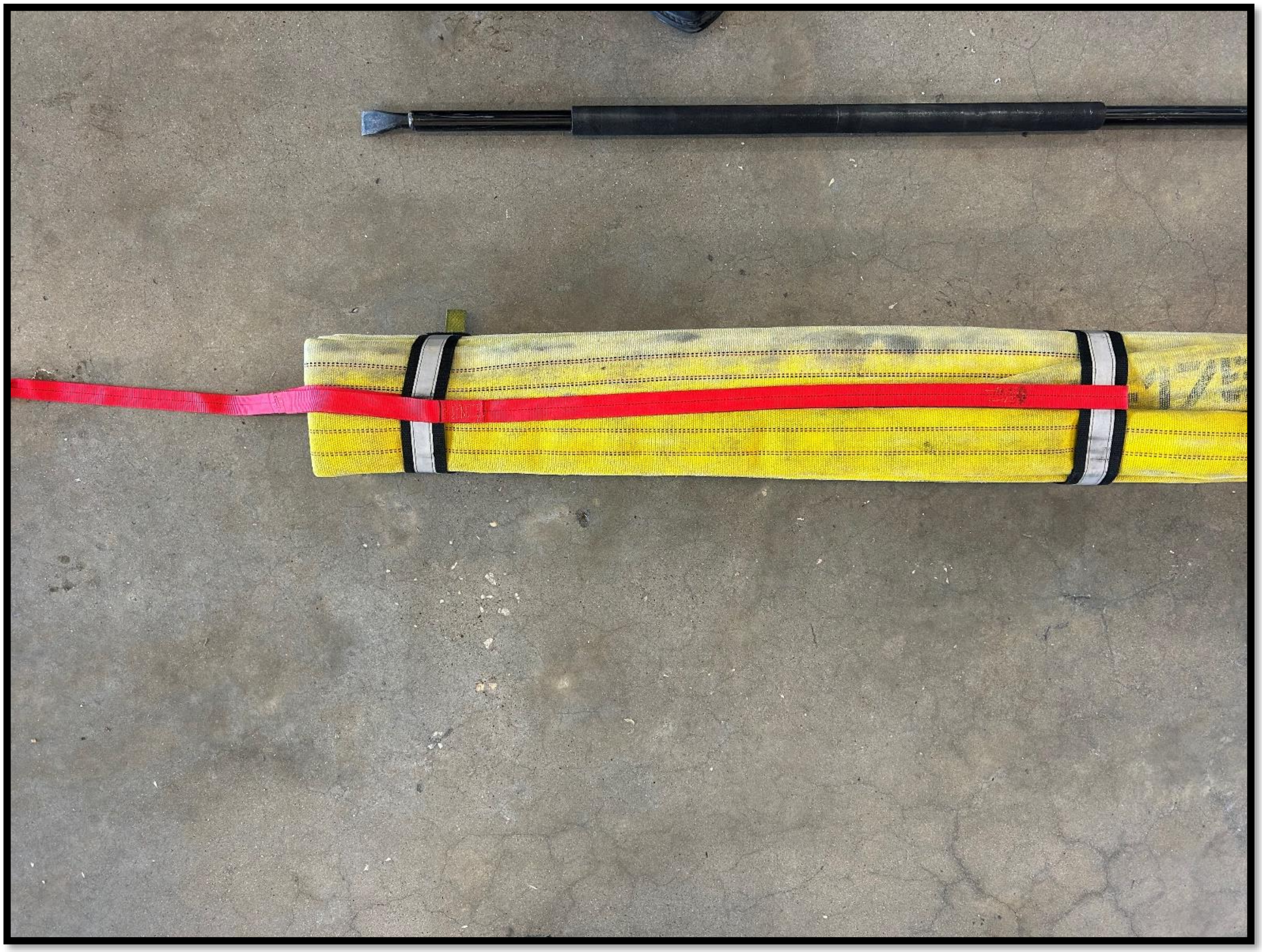


Figure 8. Bundle oriented upside down for demonstration purposes.

8. Once the bundle is secured the bundle should be loaded into the rear hose bed and connected to the applicable 2.5” supply load with the pull strap oriented on the bottom of the load and the tail end running towards the tailboard.

Rear loads:

Each apparatus is equipped with rear bed hose loads for the purpose of fire attack. The specific locations of these loads may be different on each apparatus; however, they will be loaded using the same method. In the instances where an engine is not equipped with two trunk line beds, the configuration utilizing the skid load will be utilized. The following section will describe each of the loads.

2 ½ attack line

The 2 ½ attack line will be loaded on the portion of the rear hose bed that is on the driver's side of the apparatus. It will consist of 600 feet of 2 ½" hose in a flat load and finished with a 2 ½ smooth bore nozzle. All 2 ½ smooth bore nozzles will be the department standard 1 1/8" tip with an integrated 1 3/16" tip. Note: While it is preferred that all hose be the department standard TRU-ID hose, if there are not available quantities of TRU-ID hose, the 300' that is directly preceding the nozzle will be the correct hose.

Begin the load on the left side of the hose bed (driver side) by taking the first 2 ½" coupling to the back of the bed. The first fold will be made even with the edge of the hose bed. The next fold will be directly next to the first fold, and the third fold next to the second fold. The subsequent folds will be in this fashion with all folds in line with the edge of the bed for the first 300 feet.

Once 300 feet is loaded in this manner, the next fold will be on the left most side of the bed with a short loop (approx. 9"). Continue loading on top of this first fold with all subsequent folds even with the edge of the hose bed until a stack of 100 feet is complete.

The next fold will be next to the first stack with a slightly longer loop than the first (approx. 12") In the same fashion as the first stack, load 100 feet of 2 ½".

The final stack (completing the 600" load) will begin next to the second stack, with a longer loop (approx. 18"). Continue loading the final stack with folds even with the hose bed. A 9" hand loop will be placed after loading the first section to denote the 50" mark. And the last 50' loaded to finish the 100' stack. The load will be finished with a smooth bore 2 ½" nozzle.

Skid Load

The Wye'd line will be loaded in the rear hose bed on the in-board right side of the apparatus. The load is constructed of 300' of 2 ½" and finished with a 100' hose bundle (for construction of the hose bundle see the section of this manual on 'Bundle ').

Begin the load on the left side of the divided hose bed by taking the first 2 ½" coupling to the back of the bed. The first fold will be on the left most side of the bed with a short loop (approx. 9"). Continue loading on top of this first fold with all subsequent folds even with the edge of the hose bed until a stack of 100 feet is complete.

The next fold will be next to the first stack with a slightly longer loop than the first (approx. 12") In the same fashion as the first stack, load 100 feet of 2 ½".

The final stack (completing the 300' load) will begin next to the second stack, with a longer loop (approx. 18"). Continue loading the final stack with folds even with the hose bed. A 9" hand loop will be placed after loading the first section to denote the 50" mark. And the last 50' loaded to finish the 100' stack. The load will be finished with a gated wye with the hose bundle attached.

Leader Line

On apparatus where it is possible to do so, an identical load will be in the divided hose bed to the right of the skid load (on the far-right side of the hose bed). The only difference being that in place of a gated wye, a 2 ½ smooth bore nozzle will be used to connect the 2 ½" to the hose bundle. A "leader line tip" must be used on these nozzles (a 1 1/8 tip with 1 ½" male threads at the nozzle tip.)

Supply Line

5" will be loaded in a standard supply load as follows: Begin the load on the left side of the hose bed (driver side) by taking the first 5" coupling to the back of the bed. The first fold will be made even with the edge of the hose bed. The next fold will be directly next to the first fold, and the third fold next to the second fold and so on. The subsequent folds will be in this fashion with all folds in line with the edge of the bed for the first 600 feet using 100' sections. The next section will be a (red) 50' section of 5" loaded in the same fashion as previously. Another 100' section will be loaded and then the load will be finished with a (red) 50' section. This should result in 800' of 5" hose.

<u>Section V: Hose Deployment</u>

Hose deployment will follow the same concept of operations as the hose loads. The Nozzle Firefighter will have a singular method of deployment. **<u>The use of rear hose beds in deployment will be the only PFD method of extending lines to structure fires, this allows the most amount of flexibility and function.</u>** At the discretion of the company Officer, the tactic of deploying a dry line to a tenable area outside of the immediate fire area is authorized. Crews employing this tactic will constantly monitor heat and fire conditions during the deployment and should not hesitate to retreat from the environment should conditions change. When this tactic is utilized, the Driver/Operator will suspend other fireground duties and will be standing by to immediately charge the line at the command of the interior crew. When the skid load or leader line is employed, the 2 ½” can be charged by the D/O and the Control Firefighter will open the valve at the command of the interior crew.

Rear deployment

The Nozzle Firefighter will begin by assessing the stretch and estimating the lengths of hose that will be needed. They will then place the bundle onto their shoulder. They will then reach with the opposite hand to take a fold of the supply hose and proceed to the drop point. The Nozzle Firefighter is responsible for up to 200 feet of hose (the bundle plus 100 feet of 2 ½”). Based on the stretch estimation, the Control Firefighter will assist with pulling additional loops to clear the desired amount of hose. The Control Firefighter is responsible for the deployment of the remaining 200 feet of 2 ½” if necessary. The Driver / Operator will pull an additional section of 2 ½” beyond that which the attack team estimated in order to have enough slack to make the discharge connection without interfering with the attack stretch. As an example, if the attack team estimates a 250’ stretch, 300 total feet of hose will come out of the bed. The Driver / Operator will break the coupling at the desired length and make the connection to an available discharge. It is imperative that the D/O knows for certain the total quantity and type of hose that is deployed in order to make appropriate pump calculations.

Stretch Estimation

With the use of static beds and rear deployment, there is a corresponding need to practice the skill of stretch estimation. This is a skill that is developed over time and practice must be drilled regularly. There is, however, a simple formula that is generally effective at correctly estimating the amount of hose needed for a stretch.

- Travel + width of the structure + Depth of the structure + 1 section per floor above ground.

Travel hose – This is the distance / amount of hose that is needed to travel from the rig to the door. This does not include any of the hose that will be needed in the interior for fire attack. With the increase of available hose with no notable hydraulic drawbacks, the initial arriving engine can spot their apparatus significantly further away, allowing for better placement of other apparatus. When deploying from the rear bed, this travel hose will be made up with 2 ½" hose.

Working hose – This will be the portion of hose that is inside the structure from the point of entry to the nozzle. This is estimated by adding the length and depth of the structure and rounding up to the nearest 50'. An additional 50' section will be added for every floor above the ground. If required, it is authorized for the wye or leader nozzle to advance into the structure beyond the entry point to ensure sufficient working length is available. It is imperative that this appliance is secured in the open position once charged to prevent accidental closing.

Hose handling / advancement

When utilizing these techniques, firefighters will discover they can easily manage and advance any hand line PFD utilizes. The task will not be "easy," but with good technique you will prevent immediate fatigue and possible injury. These techniques also facilitate immediate hose advancement without any repositioning.

Nozzle positions

Clamp

Begin by laying the hose line on the ground. The nozzle firefighter will position themselves with their right or left knee on the hose. The nozzle firefighter's opposing foot will be positioned in front of the firefighter with the heel resting on the ground to help balance. Knee should bend slightly wider than 90 degrees in a comfortable position. The nozzle will be positioned in front of the nozzle firefighter approximately 3-4ft from the supporting knee's contact with the hose lines. The hand opposite of the arm clamping the hose will activate the bale of nozzle and hold the hose near the coupling while the other hand will support the bend near the ground. The firefighter will support the underside of hose with thumbs pointing towards nozzle. The firefighter will sit back on the heel of the back supporting leg to create friction to combat nozzle reaction. The firefighter's hands should only be managing direction while the leg and body weight manages nozzle reaction.

Comella Grip

Begin by putting their left or right knee on ground and putting the opposing foot angled 45 to 90 degrees away from the opposing knee. Face the torso in the similar direction as the foot that is angled out. Position the hose line on the hip on the same side of the body The angled foot will keep toes in active position and centered under firefighter's weight. The nozzle will be positioned well in front of the firefighter to the point where stretching the unoccupied hand forward can reach the bale at the fingertips. A firefighter will bring the bicep of the arm holding the hose down around the hose line and use the bicep to compress the hose line into the torso. The elbow will be positioned on the inside of the thigh with palms supporting under side of hose line with thumbs pointing towards nozzle. The nozzle hand will support near the coupling and the opposing hand will support the hose where it can reach without repositioning the elbow. When opening the bale, the firefighter will squeeze the hose line into torso to combat nozzle reaction. Firefighter's hands should only be managing direction while the torso manages nozzle reaction.

Attack team advancement

While advancing a charged handline in a structure fire, the Nozzle Firefighter should pay special attention to ensuring that the method of water application is sufficient to effect a change in the environment. This will determine the speed and method of water application that is needed to cool the compartment effectively. The fire attack team will generally transition between the three movement speeds as conditions dictate.

1. Move – This is the fastest method of advancement and is used where conditions are good and little heat is present. Visibility is good and the attack team can move efficiently throughout the environment without the need to flow. The hose line can be charged or dry during this phase based on fire conditions
2. Hit and move – This phase of the attack is going to occur when the conditions are met where the attack team needs to flow water to cool the environment and begin the fire attack. The nozzle is opened from a stationary position. The Nozzle Firefighter will generally work the nozzle in an “inverted U” pattern, working high to low and near to far to contract the heat and expanding gases back toward the source. The Nozzle Firefighter will flow from a stationary position until the desired change in conditions is met. The attack team will then shut down the line and take a moment to observe conditions. If conditions remain improved this will signal the attack team that they can proceed further until they are met again with addressable fire conditions. This process will continue until the attack team makes the fire room or until conditions are such that the attack team cannot advance sufficiently without needing to flow.
3. Flow and move or “push” – This phase of attack will likely be the final phase as the attack team moves closer to the area of origin. This method of movement is needed when the line must be kept open, uninterrupted, to achieve environmental change effectively. The Nozzle Firefighter will open and work the nozzle in the same pattern and method as the hit and move but will begin to move toward the fire compartment. This is a very slow and methodical process, and the attack team must not outrun the effectiveness of the stream. The push will continue until the crew makes the fire compartment and can finally direct the hose stream onto the burning materials and complete the extinguishment. Note: once the attack

team can begin placing the hose stream in the fire compartment, they should call for “vent” from the OV team, signaling the OV team to take any intact windows in the fire compartment to allow heat and the products of extinguishment to escape the compartment.

Control position

It is the responsibility of the Control Firefighter to manage the hose in a manner that facilitates maximum GPM flow from the nozzle. This involves addressing kinks, pinch points, impending damage, and any other variables engine companies may encounter. Control is responsible for facilitating hose advancement into and throughout the structure. The goal of the control position is to be proactive and anticipate what the nozzle position may need. This includes preparing additional hose in a structure in a manner for unhindered and immediate advancement but also monitoring surroundings for firefighter-threatening conditions that could have been missed by the officer and nozzle positions.

The Control Firefighter will position themselves well behind the Nozzle Firefighter to manage slack and assist with nozzle reaction. Begin by placing the knee closest to the nozzle on the ground by the hose. Place the opposite foot extended behind you. The hand closest to the nozzle will be placed flat on the ground on the opposite side of the hose line. The hand that is furthest from the nozzle will support the underside of the hose line with the thumb facing either direction. In this position, Control can facilitate advancement by gripping the hose tightly in the hand that is furthest from the nozzle and crawling forward. The goal is to assist but not push your Nozzle. When advancement has stopped the hose should be as straight as possible behind the Nozzle to facilitate the nozzle reaction down the hose line.

Section VI: Water supply

Water supply refers to the source of the extinguishing agent used to mitigate the fire. This can range from the onboard booster tank of the first arriving engine, to complex multijurisdictional nursing and relay operations. Water supply in any form must be anticipated as much as possible to ensure that it is uninterrupted as the fire ground develops. Arriving members, particularly the initial arriving engine D/Os, need to be constantly assessing their water supply needs and continuously improving the water supply situation. This section will establish the essential elements of water supply for most incidents. D/Os and incident commanders need to be creative in their implementation of these principles within the confines of this manual.

Water supply will be the responsibility of the 2nd due Engine Company D/O. That this member is *responsible* for this task does not implicitly mean that they will be the person to secure the hydrant in all scenarios. The first 2 arriving Engine D/O's will facilitate the establishment of a reliable water supply to ensure the needed fire flow is met as quickly as possible. **Order of priority for continuous water supply will be:**

1. Hydrant connection made by second due engine company. Whenever possible this should be a reverse lay with the laying out pumper hooking up to the hydrant. When reversing to a hydrant, the water supply pumper will hook up to the hydrant and supply the attack pumper. While often this boost in pressure is not needed, this action will set up for unforeseen incident needs (primary pump failure, increased water demands, etc.) without significantly delaying water supply.
2. Nursing (stacking tanks) by 2nd due if delay for hydrant supply. The fastest method of supplying water to attack lines is the onboard tank water. This can be augmented by the 2nd due engine nursing (stacking tanks) to be able to quickly utilize 1500 gallons of water until able to obtain continuous water supply. This option should always be considered as it is the fastest and is always available regardless of the availability of hydrants. The decision to delay hydrant supply could be for several reasons. This includes but is not limited to:
    - Unavailability of reliable hydrants
    - Intentional delay to allow for proper placement of apparatus (i.e. delaying charging the supply hose to allow ladder company to position properly.)

- Intentional delay to rapidly utilize manpower to affect a rescue or rapid fire attack.

3. Relay Operations may be necessary to achieve continuous water supply operations for building fires when municipal water sources are available, but a long offset exists. This should be done by placing an engine after every 500’ of supply line to boost pressure. A few examples of this would be:
    - Storage Facilities with long driveways and hydrants at the road.
    - Industrial/Commercial facilities with long driveways and hydrants at the road.
    - Sites that have long offsets that have hydrants at the road, however, not enough ingress/egress to allow for nursing/tanker operations. (ex: narrow roads that don’t allow apparatus to pass).

Note: Delaying the establishment of a water supply for tactical reasons does not negate the need for a continuous water supply. On all incidents where sufficient hydrants are present, they will be utilized.

4. Continuous nursing operations. Nursing operations will be the preferred water supply method when municipal water supplies are not available or practical. The second due engine will still be responsible for water supply as the nurse engine.
    - Incident commanders will request additional apparatus as the incident demands. The second due apparatus will serve as the nurse pumper and be the water supply source for the attack pumper. When available and practical, a tanker should be used as the nursing apparatus.
    - Additional arriving apparatus will work to ensure that the nurse engine stays full. The nurse and attack pumper operators will be in constant communication to manage available water supply.
    - An intermediate (relay) pumper will be used every 500’ when deploying 5” supply hose. This is based on territory evaluation, past experiences, and potential calculations for water supply needs.

Drivers should try to project future water needs as the incident develops. Always improve your ability to deliver water to the incident and maximize resources. If a secondary water supply is needed, a hydrant on an opposite loop should be located and used if possible as opposed to the same loop.

Section VII: Riding, Tool, and Task Assignments

Residential Deployment

1st Due Engine Co

- Quick 360 if able
- Fire Attack – GET WATER ON THE FIRE
- Fire area search

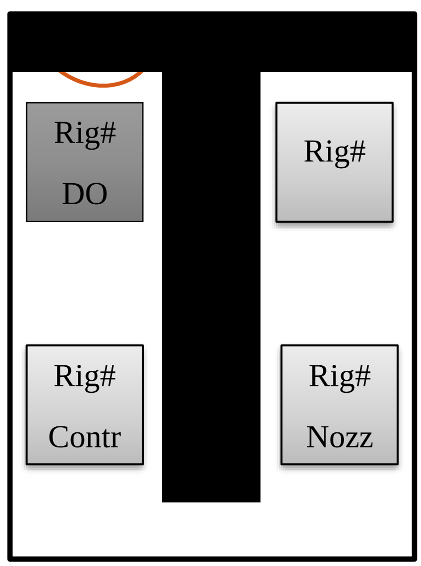
2nd Due Engine Co

- Water Supply (Lay or Hook In)
- 2nd Line/Back Up Line/Assist 1st due with line stretch
- D/O Assist 1st due Engine with water supply and other duties

3rd Due Engine Co

- RIT/On Deck – Soften the building, ladder egress
- D/O stays with crew

Engine Company Riding Locations and Radio ID



Tool Assignments

Riding Assignment	Alarms	Fires	Rescues/Accidents
Officer Radio ID Rig #	Radio – Big Light SCBA & TIC Forcible exit tool	Radio – Big Light SCBA & TIC Forcible exit tool Pike pole (Left at door)	Radio
Nozzle Radio ID Rig# Nozzle	Radio – Big Light SCBA Can Pike pole (6' Min)	Radio – Big Light SCBA Hose	Radio Rescue Tools
Control Radio ID Rig# Control	Radio – Big Light SCBA & TIC Irons	Radio – Big Light SCBA & TIC Irons Hose	Radio Vehicle Stabilization Battery cutters/wrench
Driver Radio ID Rig# D/O	Radio Check closest hydrant	Radio Water supply	Radio Power Unit Protection line Traffic safety

Section VIII: Hydrants

Dressing Hydrants

All hydrants utilized for any operations will be fully ‘dressed’ regardless of anticipated need. Fully dressed hydrants will have:

- The hydrant bag with all contents at the hydrant.
- A gated hydrant valve attached to the 2.5” discharge of the hydrant facing towards the attack pumper.
- As time and incident priorities of work allow, all engine operators involved in water supply should work to improve their water supply situation. This will typically involve filling available water tanks and adding intake hose from the hydrant to the water supply pumper. This should be done at all incidents regardless of complexity.

Hydrant identification:

- The hydrants in Pearland are color-coded by water main size. The hydrant’s bonnet color represents the water main size associated with that hydrant.
    - Red: 20”
    - Yellow: 16”-18"
    - Green: 10”-12”
    - Orange: 8”-10”
    - White: 6”
    - Blue: Unknown
        - Hydrants with a blue bonnet represent hydrants installed by private entities (e.g., Hotels) whose water mains are not determined by the municipality.

The 2nd in Engine Co. is responsible for securing a water supply. Water supply will be directed by the 1st in Engine Driver / Operator and facilitated by the 2nd due Engine Co. Driver / Operator.

Every effort should be made for the 2nd due pumper to approach the scene from the opposite direction of the 1st due pumper.

Procedure: Catching a hydrant using a reverse lay

1. D/O stops the apparatus at the fire scene near the attack pumper. Crew (minus the driver) will quickly exit and collect tools and equipment from their apparatus and close all doors and compartments.
2. The Control Firefighter then grabs the 5" hose and pulls approximately 20'-40', enough to ensure that the 5" hose will flake off when the apparatus leaves the scene. Sliding one flake of 5" under the wheel opposite the direction of travel. The first due D/O can perform this task if they are available, thus freeing the Control Firefighter to remain with their crew.
3. The firefighter shall then anchor the 5" hose to ensure that the hose flakes off the apparatus.

The supply hose will be anchored by folding the hose accordion-style against a solid surface (i.e. tire of a stationary vehicle or curb).

Procedure: Catching a hydrant using a forward Lay

1. Control Firefighter exits the apparatus and throws the hydrant bag at the hydrant.
2. Firefighter then grabs the 5" hose and pulls approximately 20'-40', enough to ensure that the 5" hose will flake off when the apparatus leaves the water supply site.
3. The firefighter shall then anchor the 5" hose against the hydrant to ensure that the hose flakes off the apparatus.
   a. The supply hose will be anchored by folding the hose accordion-style against the hydrant.
4. Once the firefighter determines that the 5" hose is secure, the firefighter will give the Driver / Operator the command to drive forward, the command shall be PREDETERMINED radio traffic or hand signals.
   a. The hand and arm signal will be the firefighter at the hydrant raising one arm above the head, palm away with fingers extended and joined, and moving the arm away the body. This motion is repeated until the apparatus begins to move, indicating that the message was received. The hand and arm signal will be the preferred method and will be applicable in almost all situations as the apparatus will be near the firefighter giving the signal.

   b. When the hand and arm signal is not practical, the radio message “ENGINE ___, LAY OUT” will be used by the firefighter at the hydrant. Because this traffic is going to be expected, no preparatory message needs to be relayed prior to this message.

5. If the 5” hose is determined not to be secure or does not flake off the back of the apparatus DO NOT ATTEMPT TO STOP THE 5” HOSE. The Control Firefighter will use the radio to tell the driver operator to “STOP”.
   a. The Control Firefighter shall continue to have visual contact with the 5” hose until one 100’ section is off the apparatus if distance permits.
6. If the 5” hose is determined to be secure and successfully flaking off the apparatus, the firefighter shall attach the 2.5” gate valve to the 2.5” connection facing the fire scene on the hydrant and the firefighter shall ensure that the gate valve is CLOSED.
   a. The hydrant wrench shall be used to remove the 2.5” connection cap to save time if the cap is difficult to remove by hand.
   b. At this time, the Control Firefighter shall ensure the other 2.5” cap is tight. Failure to perform this check can result in injury due to the 2.5” cap coming off without warning.
   c. The firefighter SHALL REMAIN BEHIND (opposite the steamer) the hydrant while opening the hydrant to flow water out any outlet.
7. Once the 2.5” gate valve is securely on the 2.5” connection and CLOSED, the Control Firefighter shall open the hydrant to flush out discolored water and or debris.
   a. To flush the hydrant, attach and secure the hydrant wrench to the stem nut of the hydrant. Then, remove the steamer outlet cap. While standing behind the hydrant, turn the hydrant wrench in a counterclockwise direction to begin the water flow.
   b. The hydrant does not need to be fully opened to flush the hydrant successfully. The firefighter shall close the hydrant when continuously clean water flows for a minimum of 3 seconds.
   c. The firefighter shall turn the hydrant wrench in a clockwise direction to stop the water flow.

8. Once the hydrant is successfully flushed and closed. The firefighter will attach the 5” hose using the Storz connection or threaded adapter. DO NOT REOPEN THE HYDRANT AT THIS TIME.
9. Using the radio or clear hand and arm signal, the Control Firefighter shall communicate with the Driver / Operator that the hydrant is ready to supply water.
    a. At this time, the Control Firefighter shall ensure that the 2.5” gate valve is in the closed position.
10. The Control Firefighter SHALL NOT open the hydrant until the Driver / Operator gives the order to do so.
11. Once the Driver / Operator gives the command to send water, the Control Firefighter shall COMPLETELY open the hydrant.
    a. It may take 19 to 21 turns to fully open the hydrant.
    b. Once the hydrant is fully opened, the Control Firefighter will leave all hydrant bag tools at the hydrant and make their way to rejoin their crew. The Control Firefighter will observe the charged supply hose while moving to the scene and report any concerns to the Driver / Operator of the attack pumper.

Section IX: Target Flows

This section will establish target handline nozzle flows for PFD for the purpose of extinguishment of common structure fires. These target flows have been determined based on internal research and development as well as research conducted by the UL Firefighter Safety Research Institute studies. These shall be the achieved flows for all offensive operations the Pearland Fire Department responds to with a reasonable variance of no more than +/- 10 GPM. All interior attack lines will be equipped with solid stream (smooth bore) nozzles with no exceptions.

Target flows and equipment:

Hose lines / fire streams selection shall flow adequate Gallons Per Minute (GPM) to sufficiently overwhelm fire production. This flow can also be considered in Gallons Per Second (GPS). Modern materials and home furnishings have been proven to have a significantly higher heat release rate (HRR) causing more rapid-fire growth and resulting in more advanced fire conditions upon fire department arrival. The equipment selected has been proven through testing to provide sufficient flow under standard operating conditions, while still providing a line that can be maneuvered effectively and safely. For offensive fire operations the following flows will be achieved:

1 ¾ handlines

- 160 GPM (2.5 GPS)

2 ½ handlines

- 1 1/8” solid stream 260 GPM (4.5 GPS)
- 1 3/16” solid stream 300 GPM (5 GPS)

NFPA defines the initial attack line as the first hose stream placed into service by a company at the scene of a fire in order to protect lives or to prevent further extension of fire while additional lines are laid and placed in position. (NFPA 1410). Because this line is designated as a life safety line, it should be uncompromised and uninterrupted coming from a single point of discharge whenever it is possible to do so. Highest tactical priority should be given to the proper selection and staffing of the initial handline, with all other lines serving in a supportive role to the initial attack line.

Nozzles that are equipped with integrated tips will not have an additional tip that has an equal or larger diameter or rated flow. The only integrated tips that should have an additional tip will be on the 2 ½' nozzle.

Any line operating for structural firefighting from a standpipe system shall have a solid stream (smooth bore) nozzle. Solid stream nozzles are less prone to blockage from standpipe debris and generally function in a predictable manner when under-pressurized. (NFPA 13E). Additionally, 2 ½" wyes should not be used to connect to standpipes due to the limited pressure and flow from a standpipe system that may be equipped with pressure reducing valves or devices (PRV / PRD). Additional lines can be connected to other standpipe connections.

Section X: Engine Company members

Engine Company Officer

Overview Of the Engine Officer

The 1st arriving engine company Officer may well have more influence on the outcome of a fire operation than any other member assigned to the incident. This Officer must possess sound decision-making skills regarding incident strategies and tactics. The choices made by this Officer will dictate the strategy and method of fire attack which ultimately impact fireground operations from arrival to extinguishment. While many of these decisions are made during the incident, the company Officer is responsible for always ensuring the company's readiness at the firehouse prior to the incident.

The primary function of the engine company Officer is to facilitate fire extinguishment by overseeing the placement of a hose line to the fire area and directly supervising its operation to extinguish the fire. To clarify, it is the engine company Officer's job to place themselves and their crew along with a hose line in a position to create and maintain savable space and protect victims.

In fulfillment of this primary function, the engine Officer is responsible to make a number of critical decisions and take decisive action. While the decisions to be made will differ depending on the specific operation, the following is an outline of actions the engine Officer should expect to make from time of alarm notification to conclusion of fireground operations.

1. Alarm/Incident Confirmation.
2. Size-Up
3. Establish Command
4. Request additional/appropriate resources (2nd Alarm upgrade/Additional Tankers)
5. Conduct 360/Provide information not obtainable upon arrival.
6. Determine attack strategy.
7. Determine/Confirm appropriate hose line is in place.
8. Begin the fire attack
9. Supervise fire extinguishment.
10. Ensure relief & readiness of assigned members.

1. Alarm/Incident Confirmation:

Upon alarm notification the Engine Officer must begin to gather information from dispatch in order to confirm and, if needed, set the initial resources in motion for the appropriate type of incident. This action may also include requesting additional resources to meet the initial incident needs. It is imperative that the Engine Officer utilize the department deployment model upon alarm notification and throughout the incident to help set the conditions for success.

Information that must be obtained, confirmed, and shared with crew members includes:

- Dispatch address
- TAC channel
- Estimated arrival sequence (ex. 1st Due, 2nd Due, 3rd Due)
- Desired engine placement (ex. Pull past the residence, stage at hydrant, reverse or forward lay)
- Location of nearest hydrant/water source for tanker shuttle
- Pertinent dispatch notes
- Pertinent preplan consideration notes.

2. Size-Up:

- The transmission of the initial arrival report (size-up) is critical and has a strong influence on the overall outcome of the incident. The 1st arriving apparatus has the responsibility to transmit an arrival report over the radio. The importance of the arrival report cannot be stressed enough and when done properly allows units not on-scene to fill information gaps.
- The first few minutes after arriving on-scene can be chaotic, stressful, and possibly require the company Officer to be physically involved in performing company-level task but none of these conditions negate the importance or requirement to transmit an arrival report. Refer to the PFD ICS for arrival report format and considerations.
- Command will be established on all incidents and can be either mobile or stationary. The decision to establish a mobile or stationary command is incident dependent and can evolve as the incident evolves.

3. Establish Command:

The first arriving Officer has several command options from which to choose when arriving at the incident, depending on the situation. The following command options define the company Officer's direct involvement in tactical activities unless otherwise stated in PFD Procedure 401 'Establishing Fireground Operations'.

- Investigative mode– Upon arrival, an incident may not have visible indicators of a significant event. These situations generally require the first-arriving Officer to investigate. The Officer can usually perform the role of initial IC as well as personally investigating the situation along with the company members. It is implied that when there is "NOTHING SHOWING" the investigative mode has been enacted.
- Fast-Attack mode– Situations that require immediate action to stabilize the incident. In these situations, the Officer shall work with the crew and perform initial IC. This might be the case when a company arrives at a room and contents fire or a fire with occupants in imminent danger. When utilizing this mode, the Officer should also verbalize the action they would be performing, such as "Fire Attack" or "Rescue." The fast-attack mode should not last more than a few minutes and end with one of the following:
    - The situation is stabilized
    - The situation is not stabilized, and the company must withdraw to establish an exterior command post
    - Command is transferred to another Officer
- Command mode– Large, complex, or rapidly evolving incidents require immediate strong, direct overall Command. In these cases, the Officer's involvement in tactical operations is less important than the command responsibility. The Officer should establish a command position in a safe and effective location until the next arriving Officer or chief assumes command from them.

4. Conduct 360 /Provide Information Not Obtainable Upon Arrival:

While various fireground tasks are simultaneously taking place such as the Nozzle Firefighter stretching a line and the Control Firefighter forcing entry, the engine Officer should obtain a 360 and transmit any pertinent information not observed upon arrival. Often, three sides of a structure can be observed when arriving on scene and the engine Officer only needs to see the last remaining side of the structure in order to formulate appropriate strategies and tactics. Obtaining a 360 gives the engine Officer vital clues as to the location of the fire, stage of fire growth, needed resources, and building construction considerations. Information that should be transmitted following a 360 include but are not limited to.

- Perceived location of fire.
- Utilities secured.
- Any egress points searched or found to be inaccessible (i.e., locked gates or a door blocked by plywood).
- Swimming pools encountered that may be a hazardous in low light situations or to masked firefighters.
- Household pets such as dogs could be a danger to personnel.
- Building construction considerations that pose a hazard.
- Lastly any time a 360 cannot be obtained it should be transmitted along with the reason the 360 could not be completed (i.e., locked gates or size of the structure).

<u>5. Determine Attack Strategy:</u>

- The 1st arriving engine Officer must determine the fire attack strategy and transmit any deviation from the Deployment Model to incoming units. The fire attack strategy is decided upon arrival or after the completion of the 360. The two fire attack strategies are "Offensive" or "Defensive.
    - <u>Offensive:</u> An offensive strategy involves taking direct action to mitigate the problem. This means an aggressive interior attack will be used because initial crews believe there is a chance that occupants may be inside the structure and conditions may be such that they could still be alive. This strategy will also be used to conserve property when conditions are safe enough to do so. The **initial** strategy taken at all fire incidents is assumed to be Offensive and does not need to be communicated unless there is a change in strategy (i.e. Defensive).
    - <u>Defensive:</u> The defensive strategy is chosen to isolate or stabilize an incident to ensure it does not get any worse. A defensive operation should be initiated when fire

conditions prevent an interior attack such as when the fire is beyond the control of hand lines.

- Because the words Offensive and Defensive can sound very similar when transmitted in chaotic situations or possibly due to radio distortion, it is recommended that the Offensive mode be transmitted with the action being taken (i.e. E-2 is stretching a 1 ¾ through the alpha door) *instead* of transmitting the word “Offensive” followed by the action taken (i.e., E-2 is in the offensive mode and stretching a 1 ¾” through the alpha door). Because victims and fire are assumed present on all incidents until confirmed otherwise, the **Offensive Mode is assumed and does not need to be transmitted.** Due to the potential of crews working in a dynamic environment that can deteriorate rapidly the quick recognition of the word “Defensive” should be a priority on the fireground and not easily confused with different modes of operation.

6. Determine /Confirm the Appropriate Hose line Is In Place:

- The engine Officer must determine /confirm the proper hose line has been selected and in place. The placement of the line should be based on information gathered during the size-up. The overriding objective is to advance to the seat of the fire as quickly and safely as possible. The responsibility rests with the Nozzle Firefighter for handline selection unless the Company Officer orders a specific selection. This must be verbalized to the Nozzle Firefighter.
- To aid the engine Officer in determining /confirming the appropriate hose line was selected, the following sections discuss the capabilities of both the 1 ¾” and 2 ½” hose lines, as well as their applicability to various situations. These guidelines are intended to assist the engine Officer in making a difficult and important decision.
    - 1 ¾” hose line is the primary attack line in the Pearland Fire Department for residential buildings.
    - The 160 GPM flowrate provided by the 1 ¾” hose line is sufficient to extinguish the majority of fires encountered.
    - When the 1 ¾” hose line is supplied with 50 psi at the 7/8” tip, the nozzle reaction is 60 lbs. This is the force felt by the Nozzle Firefighter.

- The increased speed and mobility of the 1 ¾” hose line enables the Nozzle Firefighter to more effectively operate the hose line and direct the water stream faster and more effectively.
- There exist situations where the flowrate provided by the 1 ¾” hose line may not be sufficient, and the larger flowrate provided by the 2 ½” hose line is needed. These situations include commercial and residential structure fires where heavy fire loads can and are encountered.
- When supplied with a nozzle pressure of 50 psi, the 2 ½” hose line will provide a flowrate of 260 GPM and a nozzle reaction of 99lbs. This is the force felt by the Nozzle Firefighter.
- While the elevated flowrates of the 2 ½” hose line provide increased extinguishment power, the resultant nozzle reactions reduce the maneuverability of the hose line. The advance of the line will be slower, and it may be more difficult for the Nozzle Firefighter to maneuver the stream as needed.
- There are five situations in which the use of the 1 ¾” hose line would not be appropriate and the 2 ½" hose line should be stretched: (ADULTS)
    - **Advanced fire conditions** - If the fire conditions on arrival are advanced to such a degree that the Officer feels the flow provided by the 1 ¾” line would not be sufficient, the 2 ½” hose may be used. 2 ½" hose can provide sufficient knock down power if used appropriately.
    - **Defensive position** - If a hose line is to be used from a purely defensive position, a 2 ½” hose line should be used. This includes hose lines stretched at an exterior operation, when the line will be operated exclusively from outside the building, such as from an adjoining rooftop, or from street level. This should be incident dependent in areas where adequate water supply is questionable. In a purely defensive position, total water flowing must be supported by the water source available.
    - **Unknown size or extent of the fire area** - If the size of the fire area cannot be determined by the engine Officer, a 2 ½” hose line should be used. This situation could be encountered in larger, non-typical buildings, where the size

or extent of the fire area cannot be readily determined at the onset of the operation. It must be understood while choosing a 2 ½" for flows, you will be sacrificing mobility. On these incidents, SLOW DOWN and use tactical knowledge to find easier and nearest access to fire area

- **Large, un-compartmented fire area** - If the fire area is large and is un-compartmented, a 2 ½" hose line should be used. While the Officer should use their discretion in assessing the size of the fire area, a general guideline is that a fire area over 50 feet wide can be considered "large". The area should also be "un-compartmented", which means that it largely consists of open areas and floor space. An un-compartmented fire area may also have high ceilings or directly access a potential roof vent. This may include large industrial or commercial occupancies, such as warehouses, places of worship, or McMansions. In such cases, the large, un-compartmented area will allow the reach of the stream to be unimpeded and hit the seat of fire from a distance. **2 ½" attack lines are required for all commercial fires.**
- **Tons of water** - If large amounts of water are needed whether exterior or interior, a 2 1/2" hose line is needed due to its ability to deliver the higher GPM such as in a commercial fire setting.
- **Standpipe operations** - When a hose line is stretched from a standpipe system, the 1 ¾" hose must not be used due to its high friction loss. In order to achieve a reliable and effective firefighting stream, larger hose with less friction loss must be used from a standpipe system.

Once it's confirmed that the proper hose line was selected, the engine Officer must then ensure the placement of the hose line is stretched in such a way that fire attack can reach the seat of the fire as quickly and safely as possible. The gold standard in the Pearland Fire Department for hose placement on a residential structure fire is the nozzle and first (and possibly second) coupling from the nozzle placed at the front door with the remaining hose in-line with the door as to aide in a quick deployment of the hose. Door hinges should be utilized as a guide as to which way the door opens.

7. Begin The Fire Attack:

- Before entrance into the structure, the Officer should ensure that each firefighter is properly equipped with all personal protective equipment. After ascertaining that sufficient hose has been stretched and flaked, the Nozzle Firefighter should call for water via the radio and see that the line is properly bled of trapped air, unless the determination has been made that a dry line is to be stretched. The Nozzle Firefighter will call for the line to be charged. Company Officers should utilize the TIC for doorway readings prior to/as entering the structure.
- Immediately before moving into the fire area with the hose line, the engine Officer should relay to the nozzle team information gathered while the line was being stretched. This information might include phrases such as: "two rooms, left, rear," "straight down the hall, second room on the left," or "holes in the floor, stay to your right-hand wall."
- The nozzle team must begin every interior fire attack through the door to the fire area kneeling low, near the floor, regardless of conditions. A sudden ceiling collapse, rapid self-venting or a fire driven by wind, could create a blow-torch effect at the entrance door and seriously injure any firefighter in its path. After entry is made into the fire area, the engine Officer can evaluate conditions and adjust or modify the method of advance used.
- Communication during the fire attack may be almost impossible due to the noise created by the stream striking walls, ceilings and furnishings. However, the engine Officer must monitor the radio for critical information that may affect the nozzle team. This includes ventilation delays, water supply difficulties, collapse potential, Mayday, and/or urgent transmissions.

8. Supervise Fire Extinguishment:

- During the advancement of the hose line, the engine Officer must constantly monitor the nozzle team's progress and the conditions around them. The protection afforded by bunker gear, masks, and hoods tends to insulate firefighters from the hostile fire environment which could cause members to penetrate unknowingly, into severe conditions. It is also important to mention for the company Officer does not become task saturated. Stay hands free if possible. After crew continuity, the Officer must understand that radio communication concerning the fire attack is imperative.
- As the hose line is advanced into the fire area, the engine Officer should communicate orders and directions to the nozzle team using as few words as possible. As progress is made, the

nozzle team can be encouraged with statements such as "you got it," "move in," and "good knockdown." The nozzle team should be advised of their progress and given estimates of how much fire remains to be extinguished.

- The Engine Company Officer's position when supervising the nozzle team must remain fluid. When the hose line is advanced into tight quarters and the nozzle team is forced to make turns and bends, the Officer may have to drop back on the line, switch sides, or even move ahead of the nozzle momentarily to allow for optimum nozzle positioning. This must be practiced during company-level training.
- The nozzle team is composed of the Nozzle and Control Firefighter under the leadership of the Officer. While some decision-making authority is delegated to the Nozzle Firefighter, it must be understood that any actions taken are under the strict supervision of the Officer in command of the hose line.
- Decisions that may be delegated by the engine Officer to the nozzle team include:
    - Direction of stream
    - Rate of advancement
    - Opening nozzle in an emergency
    - Partial shutdown of nozzle to reduce nozzle reaction and regain control
    - Calling for more line
    - Sweeping floors with stream
- Once the fire attack team has been able to initiate fire extinguishment on the main body of fire it is imperative that the Engine Officer transmit "water on the fire". When "water on the fire" is transmitted it is a signal for all those on the fireground that ventilation can be initiated in order to improve conditions on the fireground. Company Officers must be ready to negate ventilation if conditions do not warrant it.

9. Ensure Relief & Readiness Of Assigned Members:

- The high level of physical activity required for firefighting is well documented and the debilitating effects on firefighters must be recognized by company Officers. The engine company Officer should evaluate the members of their unit during and after the fire attack and promptly relieve individual members or request relief for the entire unit.

- To anticipate the need for relief, the Officer must remain aware of the amount of air each member of their unit has remaining. It is essential that the Officer communicates with members in order to monitor their air levels, as well as their degree of fatigue.
- In the event a member's SCBA vibralert activates (or if a member's HUD shows a rapidly flashing red light), the Officer must immediately have that member relieved and ensure their safe exit of the IDLH, as described in Training Bulletin: SCBA. Consider taking an account of the company's air supply after the knock down benchmark. Typically, this provides a slowdown in operative function and a chance to check in with the control and Nozzle Firefighters.
- When communicating the need for the company to be relieved, the Officer should be sure to find out which unit will be providing the relief. The Officer should then make verbal contact with the relief Officer, whether face-to-face or radio.

Driver / Operator

The role of the Driver / Operator is a critical role within the Pearland Fire Department. They are responsible for the safe and efficient operation of the fire apparatus with all the personnel and equipment of the company any time the apparatus leaves its quarters. On the emergency scene they are expected to operate independently without direct supervision to accomplish critical tasks that are essential to the safety of the firefighters and accomplishment of tactical objectives. Additionally, they are responsible for monitoring and communicating the building conditions for interior crews until additional units arrive. During non-emergency operations, they must always navigate the roads safely in all driving conditions as well as maintain the equipment and apparatus to ensure it's ready for service. To effectively accomplish this, the Driver / Operator must be a mature and responsible person and able to handle and manage acute stressors. Additionally, the driver must be able to take in information and technical data on the apparatus as well as make on the spot hydraulic calculations.

Responding to a Fire

Upon receiving a call for report of a fire, the Driver / Operator is to determine the best route to the reported site of the fire. Consideration should be given to the response patterns of other incoming units.

The first due engine company should strive to enter the block ahead of the first ladder company and from the same direction. This will allow for optimal apparatus positioning.

Apparatus positioning

Proper apparatus positioning is a critical component of an effective response and requires a coordinated effort between the Driver / Operator, Engine Officer, and first arriving ladder company to ensure optimal apparatus placement.

The engine has sufficient hose and can be positioned away from the fire building. Consideration must be given to ladder company response and the engine apparatus should be positioned so that it will not interfere with ladder company positioning.

Supplying Water

The Driver / Operator is responsible for supplying water to firefighting forces via hose lines and master streams and maintaining the provision of sufficient operating pressure throughout the operation. This occurs in a number of different ways:

- Supplying water to handlines.
- Supplying water to a standpipe system.
- Supplying water to a sprinkler system.
- Supplying water to a monitor or elevated waterway
- Supplying water to another pumper via a relay (nursing)

Supplying handlines

In order to properly supply a hose line, the Driver / Operator must know the number of lengths stretched, the size of the hose stretched, and the elevation to which the line is being stretched. Generally, this information is confirmed by communicating directly with the Control Firefighter.

Before pressure can be supplied to a hose line, the apparatus pump must be engaged using the following steps:

1. Place the apparatus transmission in “neutral”.
2. Engage the apparatus parking brake.

3. Move the “pump shift control” to the pump position (located in the cab).
4. Place the apparatus transmission in “drive”.

The parking brake on the apparatus must be set for the apparatus pump to be engaged.The pump will not engage if the parking brake is not set.

Once the apparatus pump is engaged, water can be supplied to a hose line using thefollowing steps:

1. Pull the “Tank to pump” lever. This will supply the pump with water from the booster tank.
2. Press the “Push to Prime” button on the pump panel if needed (this expels air from the pump system) (Figure 9.)

Figure 9

3. Press the “up” button or rotate the knob on the throttle control clockwise

Figure 10

to increase the throttle to the required pressure. (Figure 10).

4. Open the desired discharge gate to charge a hose line. (Figure 11).

Figure 11

The Driver / Operator ensures proper pressure is supplied to the handline by calculating the pressure needed to overcome the friction loss in each length of hose and the pressure loss due to the elevation of the hose line, while still providing the correct nozzle pressure "at the tip". While it is imperative that all Driver / Operators can perform these calculations accurately, "street hydraulics" can be used when pumping commonly encountered situations.

The following are the rules of thumb that govern the quick calculations involved in street hydraulics:

- add 20 psi friction loss per length of 1 ¾" hose.
- add 5 psi friction loss per length of 2 ½" hose (260 GPM).
- add 10 psi friction loss per length of 2 ½ hose (300 GPM)
- when used as a trunk line for 1 ¾", add 2.5 psi friction loss per length of 2 ½ hose.
- when using two handlines off wye on trunk line, add 10 psi per length of 2 ½ hose.
- add 5 psi per floor of elevation (one floor is roughly 10 feet).
- subtract 5 psi per floor of elevation loss below grade (one floor is roughly 10 feet).
- all smooth bore handline nozzles require 50 psi at the tip. Master streams require 80 psi.

Troubleshooting

A critical component of the Driver / Operator's job is the ability to quickly recognize and address problems that could jeopardize the supply of water to firefighting units. The following section will highlight common issues Driver / Operators can expect to encounter at an operation.

Once the Driver / Operator has properly charged a hose line, they must continually monitor the status of the line. This includes monitoring of intake pressure, as well as the pressure and flow of each line supplied. The Driver / Operator must be able to identify a problem from the pump panel that could compromise the maintenance of water in the hose line. Reports from attack crews will often be the only indication of a loss in pressure. If attack crews are reporting insufficient pressure, some possible causes can include:

- Intake pressure decreases = possible blockage, or insufficient water supply.

- Engine RPM decreases = possible kink in the hose line.
- Engine RPM increases = possible burst length.

The Driver / Operator needs to ensure the rig is receiving enough water to meet the demands of the hose lines it is supplying. The Engine will “run away from water” if it attempts to pump out more water than it is taking in. To prevent this from happening, the Driver / Operator should not allow the intake pressure to drop below 15 psi. Once 15 psi intake pressure is reached, the apparatus should be augmented with an additional water supply.

One option for augmentation is to “self-augment”, which is accomplished when an engine hooks up to the steamer with a 5” supply line and both 2 ½” outlets of a single hydrant with 2.5” supply line. This will facilitate the use of more of the possible capacity of the hydrant. Generally, however, this augmentation will be performed by the supply pumper and then sent to the attack pumper. If hydrant flow is not sufficient for fireground operations, additional / alternative water supply will be needed. Secondary water supplies should be secured from different mains whenever possible.

Control

Winstead, C. (2024, June 24). *Firefighter Responsibilities: Riding Control*.

When Firefighters think of the job responsibilities of the Control position; they may or may not be thinking to themselves that feeding the hose and leaning on the door frame or wall to keep it from falling over is the only task that needs to be accomplished on the fireground. While feeding hose to facilitate a smooth aggressive attack on the fire is one of the main priorities of the Control position; to say it is the only responsibility is a very misguided perception and ineffective use of manpower. In this text we are going to highlight some of the key components of performing as an effective Control Firefighter, starting from the beginning of the shift all the way to Incident Demobilization.

Preparation

At the beginning of every tour or assignment on an Engine Company, the first step to performing efficiently in any position is coming to work prepared with the mindset of THAT position. In this example when you are assigned to the Control position you should actively be thinking about what your responsibilities and tasks on the fireground will be and how you can best Prepare for those situations and have a plan or process for each type of incident. While we can't possibly project and plan for every incident; we can do our best by being as prepared as possible. Aside from ensuring that ALL the equipment on the Engine is in proper working order; the Control Firefighter should be paying extra special attention to the following:

- Forcible Entry Tools
- Saws
- Hydrant Tools and Adapters
- Ladders
- Hose Loads
- Nozzles
- Thermal Imaging Cameras (TICs)

At any time during the incident, the Control Firefighter may be expected to use these tools with or without help or supervision and be expected to get the job done in a safe, fast, and efficient manner. We will touch more on these throughout this text.

"The Right Stuff"

From a Team Building point of view there are certain traits, qualities, and characteristics that are desirable in a Control Firefighter:

- Experience
- Knowledge
- Determination
- Independence
- Accountability
- Discipline
- Aggressiveness
- Command Presence

The list goes on and on, however, let's discuss some of the most important traits.

Experience & Knowledge: The reason experience and knowledge are important in a Control Firefighter is that they need to have an overall understanding of fireground operations. They need to be familiar with what is going on at the end of the hose line, what the Officer is directing the Nozzle Firefighter to do and how the Control Firefighter can help facilitate the advancement of the hose line. As well as many other situations and skills that we will discuss later. It is important to note that years of service do not necessarily equal experience. Five years of experience with a department that runs one or two fires a year is not necessarily equal to five years of experience with a department that runs 50 or more fires a year. With that being said, it is important for Chief and Company Officers to be aware of the past experiences and knowledge of their crew members so that they can have the right person assigned to the right job. Now, with younger and smaller Departments this can be an unavoidable issue as you may have several members of your crew that have very minimal fireground experience and you may have no choice but to put someone with little experience and/or knowledge in that position.

Determination & Independence: The Control Firefighter needs to have determination and independence to get the job done because most of if not all the time the Control Firefighter is working away from the Nozzle and the Company Officer, they are expected to perform their job with little to no supervision. The Control Firefighter is responsible for all hose management from the nozzle to the pumper.

Accountability & Discipline: As discussed before the Control Firefighter often works independently along the hose line with little supervision or direction. Because of this, the Control Firefighter needs to be disciplined and stick to their task and not be tempted to perform other assignments and 'freelance'. When things do go wrong, the Control Firefighter needs to hold themselves accountable for their mistakes and use them as opportunities to learn and teach others, this will only help build confidence and trust with the other members of the crew because they know you are going to do the right thing even when no one is looking.

Aggressiveness: Some people believe the terms aggressive and reckless are synonymous and this simply is not the truth. Aggressiveness can be defined as 'hostile or violent behavior, it can also be described as 'determination and forcefulness' which is the definition that we are trying to apply on the fireground. As stated before, the Control Firefighter should carry out their duties and tasks with determination, however, they should also carry out their task with forcefulness, meaning with intensity. The Control Firefighter should be working with intensity to accomplish their tasks because lives and property are literally at risk if the Control Firefighter fails to perform.

Command Presence: While the Control Firefighter is not an Officer, they are ideally the more experienced or knowledgeable firefighter on the crew. With that being said, the Control Firefighter should have the courage to defend and protect their line. If the Control Firefighter recognizes that another crew is impeding or will be impeding the advancement of the line the Control Firefighter needs to address this immediately because this could turn deadly for firefighters and victims if we are unable to advance the line and put water on the fire.

No Firefighter shows up to the Firehouse on Day 1 possessing all the best traits and qualities; they are developed over time through experience, knowledge, mentorship, and training. First Due Building Fire Control Responsibilities

Size Up: When arriving at a working building fire the Control Firefighter, like any other member of the crew, should depart the engine with their assigned tools and begin sizing up the scene and

the building. They should know how to read smoke to understand what's going on inside the structure and what's going to happen to conditions once we make entry and how bad conditions will become if we don't intervene. They also should know what line the Nozzle Firefighter is pulling or is likely to pull based on established expectations of the company Officer.

Hose Deployment: We all should know the number one way to increase the chances of victim survival is by rapidly putting the right size hose, with the right flow into service and attacking the fire quickly. With that being said, the Control Firefighter should assist the Nozzle Firefighter with deploying the initial attack line. Most attack lines can be deployed off the engine by a single firefighter unless it is a very long stretch. If it is a one-firefighter deployment, the Control Firefighter should continue sizing up the building and continue completing the rest of their initial tasks and prep the building for attack line entry.

Gaining Access/Forcible Entry: While the First Line goes into service and the Company Officer is performing their 360, the Control Firefighter should be assessing how to make entry into the structure. This includes making the correct tool choices (Ex: Rotary Saws, Rabbit Tools, Bolt Cutters, and Irons depending on the type of structure and entrance encountered). The Control Firefighter should check the primary means of ingress, if the door is already open it should be controlled (not latched) until they are ready to make entry into the structure. If the door is locked, they should be looking for locks, heat, smoke, and any other signs of an IDLH environment before attempting to force the door.

Once the door is opened or forced, the Control Firefighter should immediately step back and stay low and briefly observe smoke conditions and the flow path created by opening the door. After observing conditions, if safe, the Control Firefighter should get below the smoke line with a flashlight and rapidly search for victims behind the door and around the immediate area surrounding the entrance. During this time, they should be observing the floor plan of the structure, noting obstacles, and searching for any indication of where the seat of the fire may be. Utilizing the TIC to observe the movement of heated gases may give some indication as to where the fire and heat are originating from and where it's going, this should be used to assist in the search of fire and victims, but not be relied on over traditional search methods as becoming overly focused on the TIC can lead to tunnel vision causing you to miss changes in the environment around you. Once you have finished your rapid search the door should be controlled (not latched) until they are ready to make entry with the rest of the attack team. This cuts off the flow path that was created from the door being opened and prevents unnecessary air from entering the structure and feeding the fire.

Making Entry: After finishing their rapid search and controlling the door the Control Firefighter should turn around and help the Nozzle flake out any additional hose and remove kinks. Once the Company Officer returns from his 360 and the attack team makes entry, the Control Firefighter should establish Control at the entrance and feed the hose to the rest of the attack team while still trying to control the front door to limit the amount of air fed into the structure. During this time, depending on the obstacles and the floor plan of the structure it may be necessary for the Control Firefighter to advance back and forth between pinch points to allow the Nozzle and the Company Officer to smoothly advance through the building.

Control Firefighters should be well versed in hose handling methods and how to efficiently move hose without gassing themselves (Ex: Rolling hose loops into a room to advance hose). The Control Fire Fighter should feed the hose to the attack team as they advance, it is good to feed extra hose into the building/room as long as the Control Firefighter is not creating additional obstacles or trip hazards and is laying or looping the hose in the building attack over supply. During this time the Driver / Operator may or may not assist by dragging slack up to the front door for the Control Firefighter, however, this should not be expected or relied on because it is ultimately the Control Firefighter's job to manage this task. Control should continue to move up and down the line and feed slack as needed until told otherwise by the company Officer.

Observing Conditions: While Control is hard at work assisting with the advancement of the line, they must be disciplined and not get tunnel vision. While the attacking team is making their push, Control needs to be observing conditions in the structure for any kind of change. They should be monitoring the heat and smoke inside the building as this is an indicator as to whether the attack team is improving conditions or if conditions are rapidly deteriorating and there needs to be a rapid change in tactics. They should also be watching for any changes in the flow path to ensure the fire does not get behind the attack team and cut them off from their initial means of ingress/egress. They should also be listening during smoky conditions for fire crackling, cracking, or moving of structural elements and victims.

Backing Up the Nozzle Firefighter: If called forward by the Company Officer, Control should move forward to the nozzle and assist with maneuvering the hose. Engine companies should be well versed in working together on the hose line so that movements become second nature and require little to no communication.

Extinguishment/Overhaul: Once the attack team has reached the seat of the fire and the Company Officer has determined that the seat of the fire has been extinguished, the truck company should be ready to begin pulling drywall, paneling, and any other form of interior covering to expose any hidden fire or hot spots while the Nozzle Firefighter continues attacking these exposed areas. The Control Firefighter needs to be prepared to perform these duties if the truck crew is not immediately available.

Second Due Building Fire Control Responsibilities

Water Supply/Catching a Hydrant: As stated before Control is expected to know how to do a lot of jobs. This all starts in the preparation stage with Control setting up their Hydrant Bag the way they want it, having checked all hose loads, and inspected all adapters and appliances. Upon arriving second due to a building fire the primary responsibility of Control is to catch a hydrant unless ordered otherwise. The Control Firefighter will utilize the procedures outlined in this manual to properly execute the appropriate actions to secure a water source.

Secondary Line/Backup Line: The next task that should be assigned to the second due Engine if applicable is to pull a second attack line (a line that is going to a different location than the initial attack crew) or a backup line (a line of equal or greater flow that is going to the same location as the initial attack team and backing them up for protection). All the same principles of First Due Control responsibilities are applicable if a Secondary/Backup Line is deployed.

Immediate Rescue: There may come a time as a First Due Engine that an immediate rescue is imminent, and action needs to be taken quickly. As we discussed before, we all know that the number one thing we can do to increase victim survival is to deploy the right size hose, with the right flow to the seat of the fire. The Nozzle should still be focused on putting the initial attack line in place to help protect victims and firefighters to facilitate the rescue. While the Nozzle is deploying the initial attack line, Control and the Company Officer should be focused on the rescue. This is where checking and knowing your ladders intimately is important, as you may be expected to deploy an extension ladder by yourself to facilitate the rapid rescue of a victim being threatened by rapidly deteriorating fire conditions.

Nozzle

Winstead, C. (2024a, May 28). *Riding Nozzle*. Firehouse.

Typically, this is the first job that a Firefighter will learn when they first set foot in the Firehouse as it is unequivocally the most positive influential job/task on the fireground. The importance of the Nozzle Firefighter should not be underestimated. This text will highlight some of the key components to performing as an effective Nozzle Firefighter, starting from the beginning of the shift, and leading towards incident demobilization.

Preparation

At the beginning of every tour or assignment on an Engine Company, the first step to performing efficiently in any position is coming to work prepared with the mindset of THAT position. Example: You are assigned as the Nozzle Firefighter for the tour/day, your thought process and mental preparedness should primarily be focused on fire attack. You should be thinking and asking yourself questions like: What loads are on this Engine? What nozzles are on them? How can they be applied to different types of fires and situations? Example: If we make a car fire what line am I going to stretch? How will I approach the vehicle? What if it's a commercial vehicle? What if there are victims trapped? What is my plan of action? While we can't necessarily imagine and predict every possible incident and come up with a game plan in our heads that's going to work *every single time.* The idea is that the Nozzle Firefighter should be thinking about these types of things throughout the tour and throughout their career to be better prepared for unexpected or unique situations. Aside from ensuring ALL the equipment on the Engine is checked off and in proper working order, the Nozzle Firefighter should pay special attention to the following:

- Nozzles
- Hose Loads

- Connections
- Bundles
- Water Can/Extinguishers

After the Nozzle Firefighter puts their gear on the Engine and checks their air pack, these items should be the FIRST thing checked on the Engine by the Nozzle Firefighter. The Nozzle is expected to have intimate knowledge of these items. Let's expand on this a little further.

Nozzles: This can be divided into two categories, Shutoffs and Tips.

- Tips: All tips on the Engine should be inspected and accounted for, these are the items that should be inspected.
- Check the gasket on the female connection.
- Check all threads
- Check for damage such as cracks or missing pieces.
- Check the teeth on all fog tips and make sure they move freely if applicable.
- Ensure the 2 ½" Shutoff has a Thread Protector on its 1 1/8" Tip Male Threads.

Shutoffs: All Shutoffs on the Engine should be inspected and accounted for, these are the items that should be inspected:

- Check the gasket on the female connection
- Check all threads
- Check the bale for proper operation, look inside and ensure the ball valve is opening and closing properly.
- Check for damage such as cracks or missing pieces

Hose Loads & Connections: All hose loads (Speedlays, Bumper Loads, and Hose Beds.) on the Engine should be inspected.

- Check that the hose is loaded properly in its designated area and that it is ready to be deployed.
- Check all shutoffs, tips, and thread protectors for tightness.
- Check all pre-connected lines and ensure they are connected to the discharge tightly.

- Inspect the hose bed and make sure all hose loads in the bed are loaded properly and ready to be deployed. Open the cover on the hose bed to make sure nothing has shifted over causing a pinch point in the load that could delay deployment.

Bundles: All bundles on the Engine should be inspected.

- Check that the bundle is loaded and secured properly with a strap/harness.
- Check the shut offs & the tips on the bundles
- Check all connections for tightness.

Water Can/Extinguishers: All extinguishers on the Engine should be inspected(checked).

- Check all extinguishers for any signs of damage.
- Ensure the water can has the appropriate amount of water in it and is charged with the proper level of air. Water cans should come from the manufacturer with a fill tube. This can be seen in the opening of the Water Can after removing the hose and nozzle assembly. While slowly filling the can with water, once you reach the proper water level, water will shoot up out of the fill tube, this indicates that the can has reached the proper water level. The purpose of this fill tube is to not only indicate when the can is at the proper water level but also to help you from overfilling the can with water which reduces the air pocket in the can that is needed to pressurize the can properly. Unfortunately, due to them being misplaced or damaged they are often missing from water cans so they can't be relied on.
- The Can is supposed to weigh approximately 27 lbs. and 8 oz. per the manufacturer. This can help you determine the proper amount of water to put in the can when it's missing its fill tube. Always check the can on your Engine to verify the proper weight. A fish scale is a helpful tool to measure this weight.
- Check the harness and strap for any tears or damage and ensure the strap has been adjusted to fit. Leave a little extra room when adjusting the strap so that it can fit over your bunker coat or try adjusting it with your coat on.
- Check the Dry Chem Extinguisher charge level.

- Check the CO2 Extinguisher for a seal. If the seal has been broken or if it feels light its charge may be questionable. The only way to determine if a C02 extinguisher has a proper charge is to weigh it. The C02 Extinguisher on Engine 3 is supposed to weigh approximately 32.25 lbs. and should be recharged if the weight of the extinguisher is 1 lb. 8 oz. less than that per the manufacturer. Again, the easiest way to do this would be to use a fish scale.
- Lastly, ensure all extinguishers have some form of a pin in place to protect from accidental discharge. A cotter pin is a great replacement for the pins that come standard with most extinguishers which tend to back out from movement while carrying or while being transported on an apparatus. It is also recommended that you secure the cotter pin to the can using paracord or string so that it is not lost and can be reused. If you've ever had to clean dry chem out of an apparatus compartment you know how valuable a secure pin is.

These are just a few things to check to ensure that your equipment is ready to be utilized. We will touch more on equipment throughout this text.

“The Right Stuff”

As discussed earlier, often the Nozzle position is filled by the new Firefighter in the Station or "The Probie". Some may look at this as a punishment, that they're considered "not good enough" or "not worthy" to do any other job. This couldn't be further from the truth, while there are many jobs on the fireground that need to be completed such as Control, High Side/Low Side Truck, O.V, etc. The Nozzle is without question the most important job on the fireground; sounds like a heavy weight to bare right? Let's talk about what it takes:

Receptivity: It is important as a Probie or as the Nozzle to be receptive to ideas, constructive criticism, and advice; always. To keep it simple: Be a sponge, there are likely a lot of people in your department or station that have been in your shoes; be receptive to what they have to offer. This is especially critical when you're on the line. It is **imperative** that you follow the orders and directions of your Officer; they are in that position for a reason and will ideally have experience mitigating the hazards you may encounter in a fire. They also should have better

Situational Awareness because they are able to observe what is happening around you while you may be focused on Fire Attack.

Situational Awareness: It is very easy as the Nozzle Firefighter for your focus to be stolen by the fire. Do your best to not fall in the trap of focusing solely on the flames, there are things happening all around you. You should still be observing smoke conditions, layout of the building, looking, and listening for victims, listening, and looking for signs of structural collapse all to the best of your ability. While your Officer is watching your back it is still critical for you to maintain your own Situational Awareness. With that being said, while we discussed it is imperative for you to follow the orders of your Officer; if you see something dangerous make sure you make your Officer aware, because they may not see the hazards you see, and this may trigger a change in tactics.

Courage: It doesn't need to be said that the profession we chose is dangerous and could potentially result in death or a life changing injury. This is because of the obvious hazards we are expected to and swore and oath to battle even if it may cost us our lives. It is important for the Nozzle Firefighter to have courage so that they can make that push down a hot hallway to rescue a trapped victim, because we swore that we would. Now, that does not mean that we should risk life or limb carelessly. We should make a sound risk/benefit analysis of every incident. That is what our Officers are for; however, if you see something, say something. All Firefighters should have the courage to say, "Hey that doesn't look safe." when it is a high risk/low benefit situation. Example: High risk of injury/death with no chance of victim rescue.

Aggressiveness: It is important for the Nozzle Firefighter (and the entire Fire Company) to perform their jobs aggressively meaning with determination and intensity. It takes courage and aggressiveness to make a push into a structure when everything in your brain is telling you "Danger! Stop!" We owe it to our citizens to perform aggressively, as their lives are the ones on the line.

While it's easy to sit here and say that these are all the things a Probie should already have when they show up to the Firehouse; it just isn't a reasonable expectation. Good Jakes don't just grow on trees, they are developed through training, experience, and mentorship. Make sure you're helping your Jakes grow! Now, let's talk about the responsibilities of the Nozzle

Firefighter when responding to building fires.

<u>First Due Building Fire - Nozzle Responsibilities</u>

<u>Size Up:</u> When arriving on scene of a working building fire; the Nozzle Firefighter should be sizing up the building. They should be asking questions like: What type of construction? What is the occupancy? Are victims trapped? Where is the smoke or fire coming from? Is there a vent point? What is the smoke doing and what does it look like? What is the required flow? How far of a stretch? etc. That's a lot of things to process in a short amount of time! This should become second nature as the Nozzle Firefighter gains experience/confidence through training and making jobs.

<u>Hose Deployment:</u> The number one way to increase the chances of victim survival is by rapidly putting the right size hose, with the right flow into service and attacking the fire quickly. The Nozzle Firefighter should be confident in what line should be pulled. In the City of Pearland often a single-family residential building can be extinguished with a 1 ¾" line, however if conditions merit a higher flow a 2 ½" line should be pulled as well as on all commercial fires. Remember: When in doubt about the required flow for fire attack there is nothing wrong with putting the bigger line in service. It's always better to have extra flow than not enough. A wise Firefighter only makes that mistake once! Most lines on the Engine can be deployed solely by the Nozzle Firefighter unless it is a long stretch, if a long stretch is encountered the Nozzle should enlist the help of Control to make the stretch with the 2 ½" off the rear with a hose bundle or just straight 2 ½" depending on conditions and tactics.

Once the proper line has been selected for deployment the Nozzle Firefighter should deploy the hose load, and they should end up with the nozzle and the 50' coupling at the entrance of the building with all hose laying attack over supply. Once the line has been deployed and laid out properly the signal should be given to the Driver / Operator to charge the line. Once the line has been charged look down the hose line for any kinks in the hose. If necessary, the Nozzle Firefighter can flow the nozzle, allow it to build up pressure and then slam the bale closed. This will cause pressure to be shot back down the line and will knock out most minor or moderate kinks. Any large kinks need to be chased down and any Control Firefighter worth their salt should be addressing these as they occur. Once all the major kinks have been worked out you should flow the nozzle wide open (always) to allow the Driver / Operator to set the proper Pump Discharge Pressure (PDP). Once the Driver / Operator has set the PDP they should signal for the Nozzle Firefighter to shut the nozzle (honk of the air horn, hand signals, etc.). Once your line has been fully deployed you are ready to advance the line into the building at the direction of the Officer.

Gaining Access/Forcible Entry: Generally speaking, on most residential structures the Control Firefighter is ideally capable of forcing entry into the occupancy with no assistance. However, after putting the first line into service and taking the nozzle to the door the Nozzle Firefighter may need to assist the Control Firefighter with forcing entry into the occupancy if they haven't already done so. Once the door has been forced the Control Firefighter will observe conditions and if safe they will do a quick search. If the door is forced and the attack crew is immediately met with very poor conditions such as heavy turbulent black smoke that is chugging out of an opening, fire, or 'black fire': It is time for the Nozzle to go to work. The Nozzle Firefighter should immediately begin addressing this by opening the Nozzle and applying water into this atmosphere. The Nozzle Firefighter should be working the Nozzle at a moderate speed in an inverted U and occasionally sweeping the ground for fallen embers and debris. They should be applying water evenly to all surfaces of the structure: floors, walls, ceilings etc. If you are met with extreme fire and/or smoke conditions, water damage is the least of the occupants worries. At this point the structure and more importantly any life inside the structure are at an extreme` risk. Applying water to all surfaces will not only help cool down the environment inside of the building (fire, smoke, etc.) it will also saturate the materials and furnishings inside the structure which will help reduce or eliminate their ability to ignite reducing the chance of flashover or other dangerous fire events.

Making Entry: Once the Officer has returned from their 360 the attack crew should make entry into the structure at the direction of the Officer. When you are the Nozzle Firefighter making entry into the structure or a room you want to take the path with the least resistance and pinch points as possible. One example would be if you are entering a room in which the door swings inward to the right, you should advance the nozzle to the left to reduce the chance of the hose getting hung up on the door slowing your advance. However, if the fire or victims warrant going that direction then you should take the most effective path to reach them. Conversely, if the door swings outward or towards you to the left the path of least resistance would be to advance into the room on the right to avoid bending the hose around that door causing another pinch point. Some pinch points (corners, furniture, appliances etc.) are unavoidable and will have to be mitigated with coordinated hose advancement with the assistance of the Control Firefighter.

Once you have advanced the nozzle into the structure, after encountering several pinch points the Control Firefighter will be moving up and down the line to ensure an efficient hose advancement. If at any point you aren't getting the hose you need to advance the line and it appears that you are no longer being fed any hose: this needs to be communicated to the Control Firefighter immediately. The line may have encountered another pinch point the Control Firefighter has not identified yet. It's also possible that debris or furniture has been shifted over on top of the hose impeding the advance. It's also possible that you've run out of hose, if this is the case a good Control Firefighter will notify you of the situation. In any case if you're not getting the hose you need to advance the line you should notify the Company Officer so they can contact the Control Firefighter to assess the situation. This allows you to focus on fire attack while they work out the situation.

Observing Conditions: While on the fireground or any incident scene every member of the crew is responsible for monitoring conditions and being safe. On a building fire on the initial attack crew the Driver / Operator is essentially your exterior safety, they are monitoring conditions and notifying the attack crew of any major changes in conditions. The Control Firefighter is bringing up the rear, they're monitoring conditions from the point of ingress all the way up the hose line. The Company Officer is watching conditions all around you and watching your six. Their job is to keep an eye on what's happening around you and to help guide your attack. However, it is still imperative that the Nozzle Firefighter not get tunnel vision and still be aware of their surroundings. The Nozzle Firefighter is typically the first person to see what the fire is doing inside the structure prior to fire attack this is important information to take note of because it can give you an idea of the general area of origin, what's feeding it, where it's going and what you need to do to cut it off from advancing to any unburned areas. This information is also important post incident once Fire Investigators begin their investigation. An example would be you found a gas can in the living room, or you noticed arcing or popping or oddly colored flames during fire attack. This information is helpful because it paints a picture for investigators as to how the fire or the building presented prior to intervention by Firefighters.

Backup Man: Once the Nozzle has reached the seat of the fire and/or more hose does not need to be fed. The Company Officer will call up the Control Firefighter to come assist the Nozzle Firefighter during fire attack to help maneuver the line making the Nozzle Firefighter's job of fire attack much easier. It is important for Engine Companies to be well versed in hose handling and working together. This will allow for a smooth hose advancement that requires little to no communication between members. This is typically developed over time through experience and training together. When an Engine Company has five members, one of the firefighters will be designated as backup.

Extinguishment/Overhaul: Once the main body of fire has been knocked down the Control Firefighter will likely move up and bring hooks to pull ceiling to reveal any fire or hot spots that may be hiding behind any interior coverings. If a drop ceiling is in place, a smooth bore nozzle can make quick work of knocking ceiling tiles out of their frames revealing fire and hot spots in the void space above. The Nozzle Firefighter should work closely with the Company Officer equipped with a TIC and the Control Firefighter pulling ceiling to mop up any remaining fire.

Immediate Rescue: There may come a time as a First Due Engine that an immediate rescue is imminent, and action needs to be taken quickly. As we discussed before, we all know that the number one thing we can do to increase victim survival is to deploy the proper size hose with the correct flow to the seat of the fire. It's easy and common to get tunnel vision when you hear those words ‘victims reported to be trapped’, however as the Nozzle Firefighter you must be disciplined and know your job on the fireground. The Nozzle Firefighter should still be focused on putting the initial attack line in place to facilitate the rapid rescue of a victim being threatened by rapidly deteriorating fire conditions. While the Nozzle Firefighter is putting the first line into service the Company Officer and the Control Firefighter can focus on how to make that rescue, your job is to protect them and most importantly the victim. While there are likely multiple examples out there to prove this point; one incident that comes to mind is the Legends on Lake Highlands Apartment Homes Fire in Dallas, TX on October 5th, 2021. Based off the limited material and media that has been released: A Fire Company arrived on scene of a two-story multi-family residential structure with fire coming from the 1st floor patio and victims trapped on the 2nd floor balcony above the fire apartment. Firefighters threw a ladder to rescue the victims and one Firefighter ascended the ladder to facilitate the rescue. The Firefighter and victims were quickly overcome by fire extending from the 1st floor and were forced to bail off the ladder. If an initial attack line had been simultaneously deployed and fire suppression started; this may have helped protect crews and victims while they made the rescue. It should be noted however, that this is an opinion based off limited information and in no way is meant as trying to take away from or demean the actions of the firefighters involved in this incident.

Second Due Building Fire - Nozzle Responsibilities

Second Due Engine Companies are assigned to establish a water supply and/or nurse the initial attack pumper as well as doing one of four things: deploying a secondary line, deploying a backup line, assisting the initial attack crew, or primary search. Below we’ll discuss some of the key responsibilities of the Nozzle Firefighter on second due assignments of a building fire.

Secondary Line: When an Engine Company arrives for a second due assignment, and they are given the order to pull a secondary line this means that the Engine Company is going to pull a second line off of the initial attack pumper or if the Engine Company is deployed in a different geographic location (such as on a large commercial structure) they may be pulling the ‘second’ attack line off of their own engine. It is typically ordered when there appears to be multiple points of origin in a structure in which the initial attack crew can’t put a stop on quickly because it's growing and moving in multiple areas of the structure, or simply because the fire area is so large, or it may be ordered when a line needs to be deployed to the exterior of a structure to protect an exposure from fire and heat impingement coming from the main structure. The key thing to remember about the 'Secondary Line' is that it is going in a different location than the initial attack line or to provide additional GPM to the total fire flow.

Backup Line: When an Engine Company arrives for a second due assignment, and they are given the order to pull a backup line this means that the Engine Company is going to pull another line of equal or greater flow from the initial attack pumper. This 'Backup Line' will be deployed to the same location as the initial attack line and the Engine Company will be tasked with assisting with fire attack due to inadequate flow of the initial attack line and/or it may be deployed to help protect the initial attack crew from being cut off from behind by rapidly developing fire conditions.

Assist Initial Attack Crew: If the initial attack crew is faced with a long stretch and/or is attempting to advance a 2 ½" line inside a commercial structure; the Second Due Engine Company may be ordered to assist the initial attack crew maneuver the line inside the structure to make an attack. Typically, this will happen if the initial attack crew has adequate flow but need help maneuvering the larger heavier hose inside a complicated floor plan. Assisting with the advancement of this hose line in this situation will be much more effective than attempting to deploy an additional line; because the sooner the initial attack crew can reach the seat of the fire the sooner, we'll be able to have a positive effect on the fire.

Primary Search: If none of these tasks are ordered to be completed the Engine Company may be assigned to perform a primary search for victims. However, it should be noted that when advancing a line or assisting the initial attack crew on the interior, a proactive Engine Company can simultaneously begin performing Primary Search especially on smaller residential structures.

Third Due Building Fire - Nozzle Responsibilities

When assigned to an Engine Company and arriving third due to a building fire, the Engine Company should be assigned to RIT (Rapid Intervention Team). The responsibility of RIT is to compile the RIT cache, soften the building (force doors, remove burglar bars etc.), throw ladders for ingress/egress, and standby for rapid deployment for Firefighter Rescue.

Section XI: High-Rise and Standpipe Operations

This section will cover Engine Company Operations in High-Rise Structures & Stand-Pipe Operations.

Definitions

Below are several definitions that are relative to High-Rise and Stand-Pipe Operations.

Mid-Rise Building – The Pearland Fire Department considers any structure that is taller than three (3) stories and less than Seven (7) stories to be a Mid-Rise Building for tactical purposes. A building with internal water supply (standpipes) and accessible by aerial ladder.

Example: Multi-Family Dwellings, Hotels, Office Buildings.

High-Rise Building – The Pearland Fire Department considers any structure that is seven (7) stories or greater to be a High-Rise Building for tactical purposes. *Example: Large Hotels, Office Buildings and Mutual Aid Structures.*

Attack Stairwell – The Attack Stairwell is the stairwell that has been selected by the Attack Team Officer as the stairwell from which the engine crew will make the attack and/or stretch.

Evacuation Stairwell – The Evacuation Stairwell is any other stairwell that is not being used for fire attack or ventilation. This will be the designated stairwell for victims to evacuate the building. If possible, occupants should be alerted to which stairwell is the evacuation stairwell if the building is equipped with a PA system.

Ventilation Stairwell – Some buildings have stairwells that are designated as ‘Smoke Towers’ that are designed to help eject smoke from the building using the building’s ventilation system.This can also be dual purposed as the Attack Stairwell as the Attack Team will introduce smoke into the attack stairwell upon entry.

Planning the Response to a High-Rise Incident

Much like your ‘bread and butter’ residential fires; when we are dispatched to a high-rise fire, we are already behind the ball on making a quick and effective fire attack. Additionally, due to the area that needs to be covered and the different tactics that are required at high-rise fires there is a much greater reflex time to getting water on the fire; this could be greater than twenty minutes depending on the height of the structure and what division the fire is located on. It is important that companies be familiar with high-rise buildings in their response area and pre-plan accordingly. Additionally, it is important to be familiar with the characteristics of stand-pipe systems, how to operate them and how to overcome possible issues:

Overview of Standpipe Systems

In the City of Pearland and it’s ETJ you may find standpipe systems in the following locations:

1. High-Rise Buildings
2. Mid-Rise Buildings
3. Hospitals
4. Warehouses
5. Parking Garages
6. Stadiums
7. Schools

Standpipe Classifications

There are three classes of Standpipes:

1. Class I – Designed for Fire Department use only, equipped with 2 ½” outlets only.
2. Class II- Designed for occupant use only, equipped with 1 ½” outlets and typically features occupant use 1 ½” hose. These systems are not designed to supply the flow required for firefighting operations and **should not** be used by fire department personnel.
3. Class III—Designed for Fire Department and Occupant use, equipped with 2 ½” and 1 ½” outlets. This system may also feature occupant use 1 ½” hose that **should not** be used by fire department personnel.

Types of Standpipe Systems

- Standpipe systems can be “wet” or “dry”. Wet systems always have water in the standpipe, while dry systems do not.
- Standpipe systems can also be “automatic”, or “manual” below are some examples of these systems:
  a. Automatic Wet Systems – Capable of providing a pressurized water supply usually using a gravity tank or fire pump.
  b. Manual Wet Systems – This system can maintain water levels inside the system however, it is unable to supply a pressurized water source.
  c. Automatic Dry Systems – These are typically supplied by a public water main and are maintained with pressurized air in the system. Once the system is activated by fire department personnel, the system will detect a reduction in pressure in the system and water will automatically be supplied to the system.
  d. Manual Dry Systems – May or may not have a dedicated water supply. If it has a water supply, it will fill with water once the control valve is opened. If no water supply is present the system will remain dry until the system FDC is supplied by fire department personnel.
- Combination Systems – Systems that supply an automatic sprinkler system and the standpipe system.

Components of Standpipe Systems

This is a brief overview of some important components of a standpipe system:

- Fire Department Connections (FDC)
  - The FDC is a 2 ½” connection or 5” Storz connection used to supply the standpipe, sprinkler, or combination system.
- Post Indicator Valves (PIVs)
  - These valves exist in systems where multiple buildings are running off the same system. The PIV is used to isolate specific buildings from the water supply. They

indicate their position with the word “OPEN” or “SHUT”. PIVs should be in the “OPEN” position during normal operations.

- Pressure Restricting Devices
    - These are devices installed at the standpipe outlet to reduce water pressure coming from the outlet. PRDs are removeable and adjustable and should not be confused with Pressure Reducing Valves (PRVs) which are not. While we’re not as likely to run into these in our district, it is important to know how to identify and overcome them. Some examples of PRDs are:
        - Orifice Plate – These resemble a large metal washer that is over the outlet opening. This can be removed with a screwdriver or other prying tool by placing it inside the orifice and prying it out.
        - Mechanical PRD – These devices look like other fire department adapters such as double females. These devices can simply be unscrewed from the hose outlet.
        - Limiting PRD – This device is located directly on the outlet valve assembly and limits how much the valve can be opened; it can best be described as having a collar and a hook attached the valve wheel that catches the collar and prevents the valve from opening further, these devices can typically be broken off without damaging the valve.
            - Another version of a Limiting PRD resembles what looks like a metal plunger sticking out front the valve wheel, this PRD can be removed by removing two set screws with Allen wrench.
            - Another version is a metal pin that prevents a collar from being turned, simply pulling the pin out will allow the valve to be opened completely.
- Pressure Reducing Valves (PRVs)
    - These valves typically exist in very tall buildings where a minimum pressure of 100 psi is required at the highest most distant outlet. Because of this PRVs are installed on all the standpipe outlets to prevent the Pressure from exceeding a set

amount on each floor. However, these can exist on any standpipe systems anywhere and should be looked out for while evaluating a standpipe for use. Here are a few key considerations when dealing with PRVs:

- There are two types of PRVs:
  - Factory preset, not adjustable.
    - These valves have their pressure set by the manufacturer and you cannot adjust them.
  - Field Adjustable.
    - These valves can be adjusted typically by using special tools designed by the manufacturer of the PRV.
      - A few examples of a Field Adjustable PRV are:
        - Zurn PRV
        - Elkhart Brass PRV
        - Giacomini PRV
  - Additional considerations for PRVs:
    - For all PRVs, but especially nonadjustable PRVs, it is important to know that the pressures are preset at the factory before being shipped to the construction site. These valves are meant to be installed on a specific outlet on a specific floor, because of human error these valves could be installed on the wrong outlet on the wrong floor and can severely restrict your flow. If you encounter a reduction in pressure and have ruled out all other possibilities such as an open valve below the operating valve or inadequate pump pressure, then you must consider this a possibility and utilize a different valve.
    - It is possible for a PRV to exist in the standpipe system and be located away from the hose outlet such as at the ceiling level.

        - Lastly, some PRVs can be hard to identify and if all else fails you can identify a PRV by taking the cap off the hose outlet and looking inside. If you can see a threaded stem, then it is likely a regular hose valve. If a smooth stem is observed, then it is likely a PRV.

- Fire Pumps
    - Many Standpipe Systems are equipped with a fire pump however, it is also common to find a standpipe system with no fire pump.
    - Fire pumps are designed to supply pressure as needed. Some fire pumps can be operated manually while others are automatic. Companies should be familiar with which buildings have fire pump systems.
    - It should be noted that once the Control Firefighter supplies the proper pressure to the handline, the Fire Pump may automatically activate and begin supplying additional pressure to the system. The Control Firefighter should be cognizant of this and adjust the outlet pressure as necessary.
- Floor Standpipe Outlets
    - Floor outlets can be in several different areas including:
        - Stairwells
        - Interior Hallways
        - Open Areas such as Parking Garages.
    - In most circumstances the floor outlet located in a stairwell is preferable to an interior hallway outlet. This allows firefighters to hook up as far from the IDLH as reasonably possible and creates a place of refuge for firefighters to retreat to in the event of any emergency by following the hose line out to the stairwell. If the compartment just beyond the stairwell door is involved, companies will be forced to make a stairwell hookup. This will be discussed in more detail later in this section.

Supplying Standpipe Systems

- Standpipe systems are supplied with 2 ½” hose or 5” depending on the system design.
- Standpipe systems can be supplied by the FDC or by a floor outlet. The FDC should be the first choice for supplying the standpipe, however, the floor outlet is another choice. This should be considered when:
    - The FDC is inoperable (due to disrepair, damage, or vandalism.)
    - It is necessary to augment the standpipe system and there is no available FDC.
- In buildings that have both sprinkler systems and standpipe systems, the first supply line should go to supplying the standpipe system.
- The third-due Engine should establish a secondary water source *preferably from a different main* and hook up to an available FDC and be prepared to augment the water supply *if needed.*

Supplying Standpipe Floor Outlets

- If the standpipe needs to be supplied via a floor outlet it should be done with 2 ½” hose in combination with a 2 ½” double female.
- If a PRD is found on the floor outlet it should be removed from the standpipe valve.
- When supplying water to a floor outlet in a wet system it is imperative to keep the outlet valve closed until water has been supplied to the outlet by the pumper. If the system is already pressured by a fire pump or other pressurized source this will cause a large volume of water to exit the outlet which will significantly affect pressure being supplied to the upper floors where the attack crew may be working.
- It should be noted that if the Standpipe Outlet has a PRV in placed, the pump operator will not be able to augment the standpipe system through this valve because the PRV will prevent backflow into the system.

<u>Pump Operations</u>

- A good guideline for supplying an FDC is as follows:
  - Supply a base pressure of 100 psi.
    - 80 psi provided at the outlet being used.
    - 20 psi friction loss in the standpipe system.
  - Add 5 psi for every floor (or every 12 ft.) above the ground floor.

Due to the unknown condition of internal pipes and fittings of the standpipe system, the system should not be charged abruptly, rather, the system should initially be charged at idle pressure and then gradually pressurized to the appropriate pressure. A sudden introduction of water, especially under pressure, could cause a failure of one or more of the components of the standpipe system.

Interior Operations and Tactical Considerations

<u>Stretching Hose Lines from the Engine</u>

A building fire occurring in a high-rise should not necessarily mandate the use of standpipes for fire attack. The Engine Company Officer may choose to stretch from the back of the engine as opposed to using the standpipe. This should be considered if the following conditions exist:

- The Standpipe System operability is in question.
- The Engine can spot their apparatus close to the building to deploy off the apparatus.
- Proximity of attack stairs to the point of entry of the building.
- The stretch is not a great distance and has few obstacles.

Additionally, here are some examples of where this may be appropriate:

- The Engine can spot the apparatus close to the building, the attack stairs are close to the point of entry and/or the fire is located on the first or second floor, this is a short enough stretch that it may be more reliable, efficient and faster to stretch from the apparatus as opposed to trying to hook up to the standpipe system.

- The Standpipe System is inoperable, and crews are forced to stretch from the apparatus.

If the Engine Company Officer decides to stretch from the apparatus, they should inform the second due Engine Company Officer of their change of tactics.

<u>Selecting the Attack Stairway</u>

All attack lines should be stretched from the 'attack stairway' and all access to the fire floor should be made from the attack stairway. The reason for this is to keep all other stairways which are 'evacuation stairways' clear of smoke or any other IDLH environment.

The selection of the attack stairwell should be made by the Engine Company Officer, when determining which stairwell will be the attack stairwell the following should be considered:

- Proximity of the stairway to the fire area on the fire floor.
- Proximity of the stairway to the standpipe outlet.
- Type of stairway used.
    - Example: Scissor, Return etc.
- Conditions on the fire floor.
- Ventilation, some buildings have stairwells that are 'smoke towers' that allow for the ventilation of smoke from the building. This should also be determined during pre-plans as well.

The Engine Company Officer should be sure of the location of the fire before deciding on an attack stairway.

There are several types of stairways that will have different impacts on the stretch:

- Enclosed Stairwells
    - Should be the preferred type of stairs as the movement of smoke is more restricted in these stairwells.

  - If there is only one enclosed stairwell and all other stairwells are open, the one enclosed stairwell should be designated the evacuation stairwell.
- Open Stairwells
  - Should be avoided for use as attack stairwells as they allow for easy movement of smoke to the floors above the fire floor. This is a problem because an unknowing occupant may use the attack stairwell to evacuate and may quickly become overcome by smoke and heat.
- Return-Type Stairs
  - As they are named, return type stairs always return to the same side of the stairwell at each floor. Example: If the door is located on the east side of the stairwell on Division 2 it will be located on the east side of the stair well on Divion 3 and so on. This is beneficial as an attack crew can assess the layout on the floor below and have a general idea of the path their hose line may take on the fire floor.
- Scissor Stairs
  - Scissor Stairs may exist in buildings where the stairwell is in the core of the building. This design has two alternating stairwells that are closed off from one another that serves each floor. Example: Stair ‘A’ begins on Division One on the West Side of the Stairwell, once you ascend one flight of stairs you are still in Stair ‘A’ but you are now on the East Side of the Stairwell on Division Two. This will continue to alternate until you reach the top floor of the building.
  - This is important to recognize because your standpipe outlets will only be located on every other floor due to the alternating stairs. This means that if your standpipe outlet and the fire are located on the same side of the stairwell you will not be able to hookup to the standpipe directly below the fire floor and make your attack as you normally would. Example: If the Standpipe Outlet and Fire are on the West Side of the building and you hook up to the outlet and advance up the stairs, you are now on the east side of the fire floor not the west side where the fire is located.

    - There are two ways to overcome this:
        - Hook up to a standpipe outlet two floors below the fire floor.
            - This is the preferred alternative as it will likely be a shorter stretch than the below tactic.
        - Hook up to the standpipe outlet on the floor below the fire floor and stretch your hose line through the floor below the fire floor and up the opposite side of the stairwell.
            - This may be even more difficult due to the occupancy layout; however, it may be your only option if the standpipe outlet two floors below the fire floor is not functioning.
        - In either situation the Engine Company Officer should coordinate with the second due and possibly the third due Engine Company as a long stretch is going to be necessary.

<u>Estimating the Stretch</u>

- When estimating the standpipe stretch the engine company Officer must determine if the standard two length is an adequate amount of hose to make the fire area.
- It's important to remember when estimating the stretch that two lengths should be in reserve to make the fire area or fire apartment.
- The estimation should be the distance traveled from the standpipe outlet to the fire area or fire apartment door plus the working length.
- As stated before, if the engine company Officer determines that more than two lengths is required, they should coordinate with the second and possibly third due engine to help extend the stretch.

Stretching the 1st Hose Line

The first due and second engine should work together to stretch the first hose line. This is ideal but will not always be the case, especially if the next due engine is delayed or coming from the other side of the city. In this case the first due engine company should begin the stretch on their own. There are at least two stretches that should be considered for Standpipe Operations: The Stairwell Stretch and Apartment Stretch

The Stairwell Stretch should be performed when the Fire on the Fire Floor has escaped its compartment and is now in the public hallway or main area of the floor causing there to be an IDLH environment present as soon as you enter the Fire Floor. This means that the Attack Engine will have to stretch inside the stairwell and begin their attack from the fire floor door.

The Apartment Stretch should be performed when the Fire on the Fire floor has not escaped its compartment such as in an individual apartment where the door is intact or can be controlled by the first unit to make the Fire Floor whether that be the Engine or Truck Company. This means that the Engine Company can stretch their attack pack dry on the fire floor to the fire compartment or fire apartment and then call for water as opposed to having to push from the stairwell.

Alternatives to Standpipes

There may be situations where the Attack Team has identified that the Standpipe System is completely OOS and is not usable. It is important for the Engine Company to have a quick reflex time when it comes to alternatives to utilizing the standpipe. Below are a few tactics that should be considered.

Well Stretch

A well stretch is where the Engine Company stretches a dry 2.5" line through the middle or 'well' of an open stairwell.

Stair Stretch

A stair stretch is where the Engine Company stretches a dry 2.5" line up the stairs until they reach the floor below the fire floor.

Hoisting Stretch

A hoisting stretch is where the Engine Company utilizes a balcony, or a window and hoists dry 2.5” hose to the floor below the fire to be used as a supply line.

Standpipe Equipment

The Pearland Fire Department Standpipe Equipment is made up of the following items:

- Standpipe Kit
  - 2.5” x 2.5” Gate Valve
  - 2.5” x 2.5” In-Line Analog Pressure Gauge
  - 2.5” x 2.5” 60-degree elbow with drain valve.
  - 2.5” Lightweight Cap.
  - Fast Wrench
  - Wedges x5
- Attack Hose Pack
  - 50’ of 2.5” TRU-ID hose in a Denver Load.
  - Elkhart 1 3/16” Shutoff with 1 1/8” tip
- Supply Hose Pack
  - 50’ of 2.5” TRU-ID hose in a Denver Load.
- Rope Jug
  - 100’ of Utility Rope
  - Large Carabiner x2

High-Rise Hose Loads

High-Rise Packs (Denver Load)

The Denver Load is the approved load for High-Rise and Standpipe Operations in the Pearland Fire Department. The Denver load is loaded in the following manner:

Supply Pack

Step 1: The load begins with the female coupling placed in front of the firefighter, making the first fold approximately 32 inches from the start of the coupling and running the hose to the left. (Figure 12.)

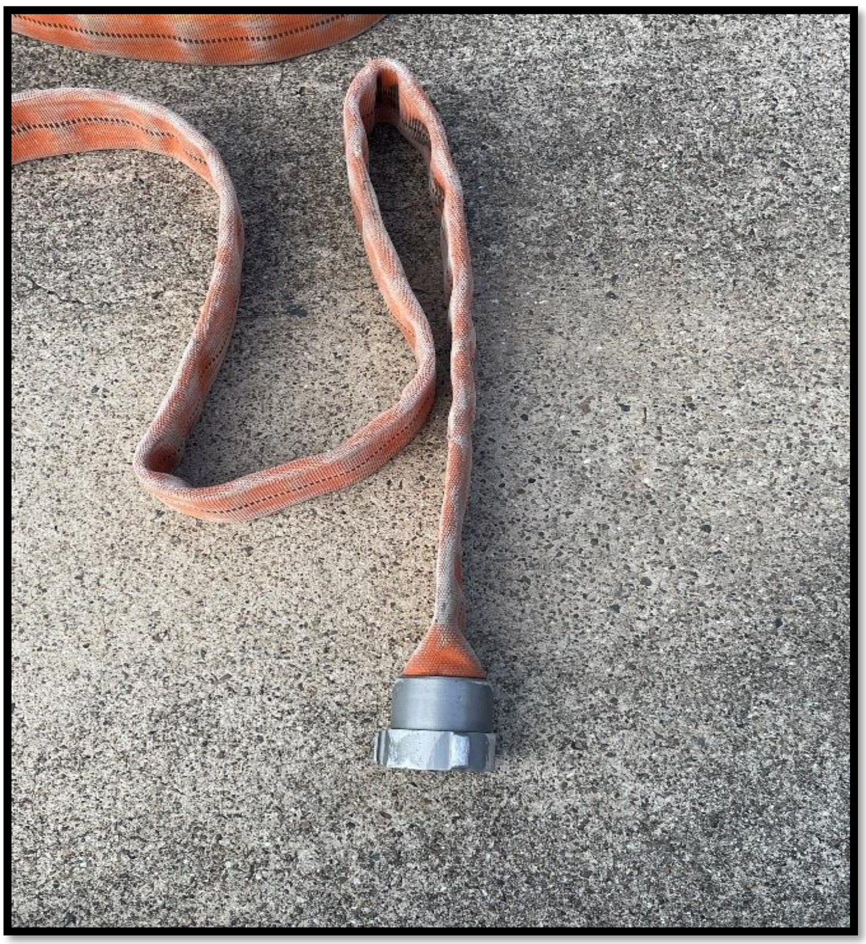


Figure 12

Step 2: The next fold will be just short of the coupling; the hose will be run to the other side of the coupling. (Figure 13.)

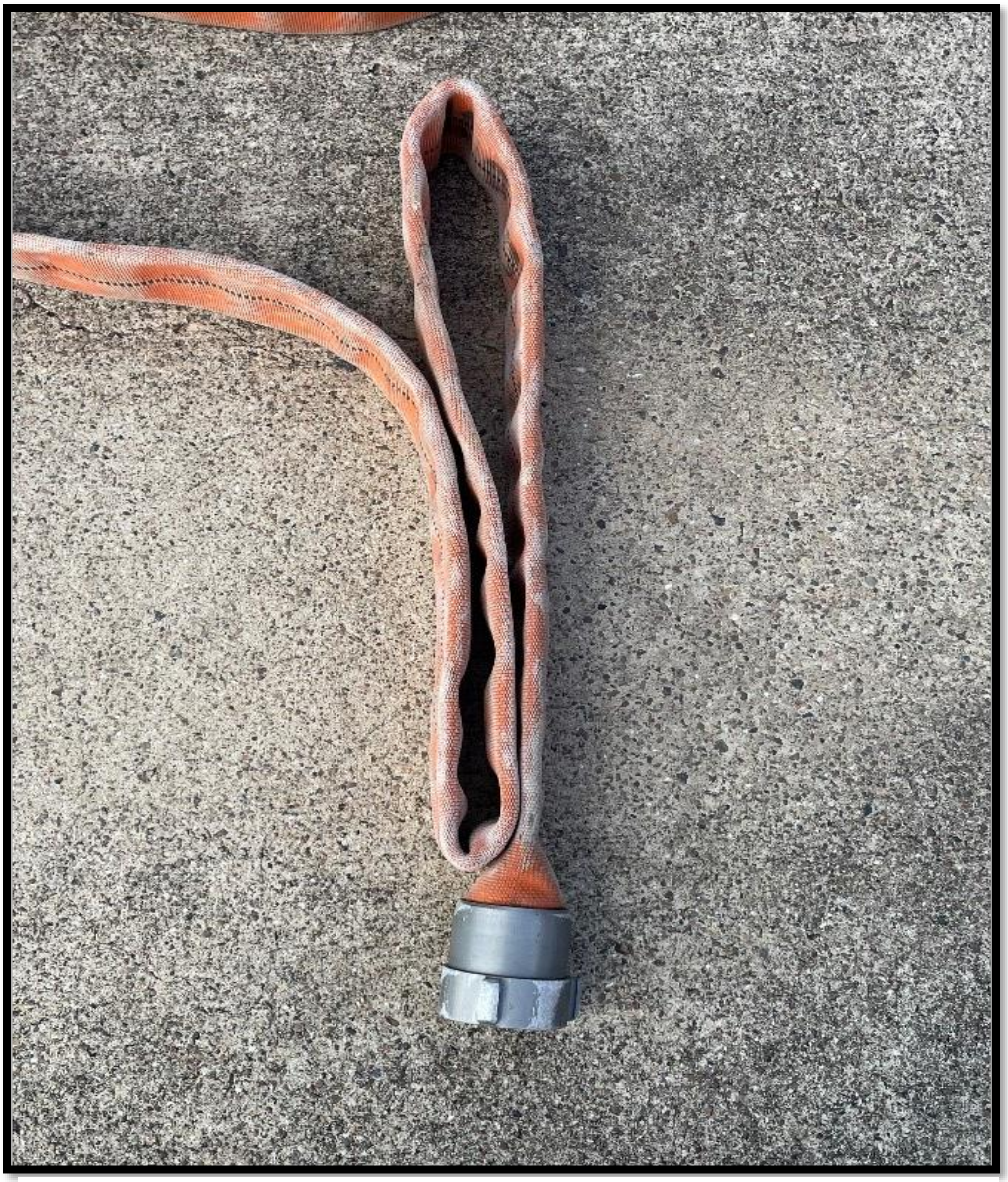


Figure 13

Step 3: Another fold just short of the coupling will be made on the opposite side. (Figure 14.)

Figure 14

Step 4: The hose will now be run to the other side of the coupling and a slightly longer loop will be made. (Figure 15.)

Figure 15

Step 5: You will then run the hose to the other side of the coupling and make a slightly longer loop, this will continue short-long, short-long until you have reached the end of the hose. (Figure 16).

Figure 16

Step 6: Once you have reached the end of the hose the female and male coupling should be on the opposite sides of the bundle. Then you will connect the male and female coupling together. (Figure 17.)

Figure 17

Figure 17

Step 7: The bundle will then be placed on edge with the male coupling on top. (Figure 17.)

Step 8: Two straps will then be placed on the pack through the innermost loop, securing the male side of the pack. It is important to make sure you orient all your straps on the same side of the pack. (Figure 18.)

Figure 19

Step 9: The bundle should be flipped and placed on its other edge and a third strap should be placed through the innermost loop on the female side of the pack. (Figure 19.)

This completes the supply pack assembly, note that all straps are oriented on the same side of the pack and can all be removed without picking the pack up. (Figure 20.)

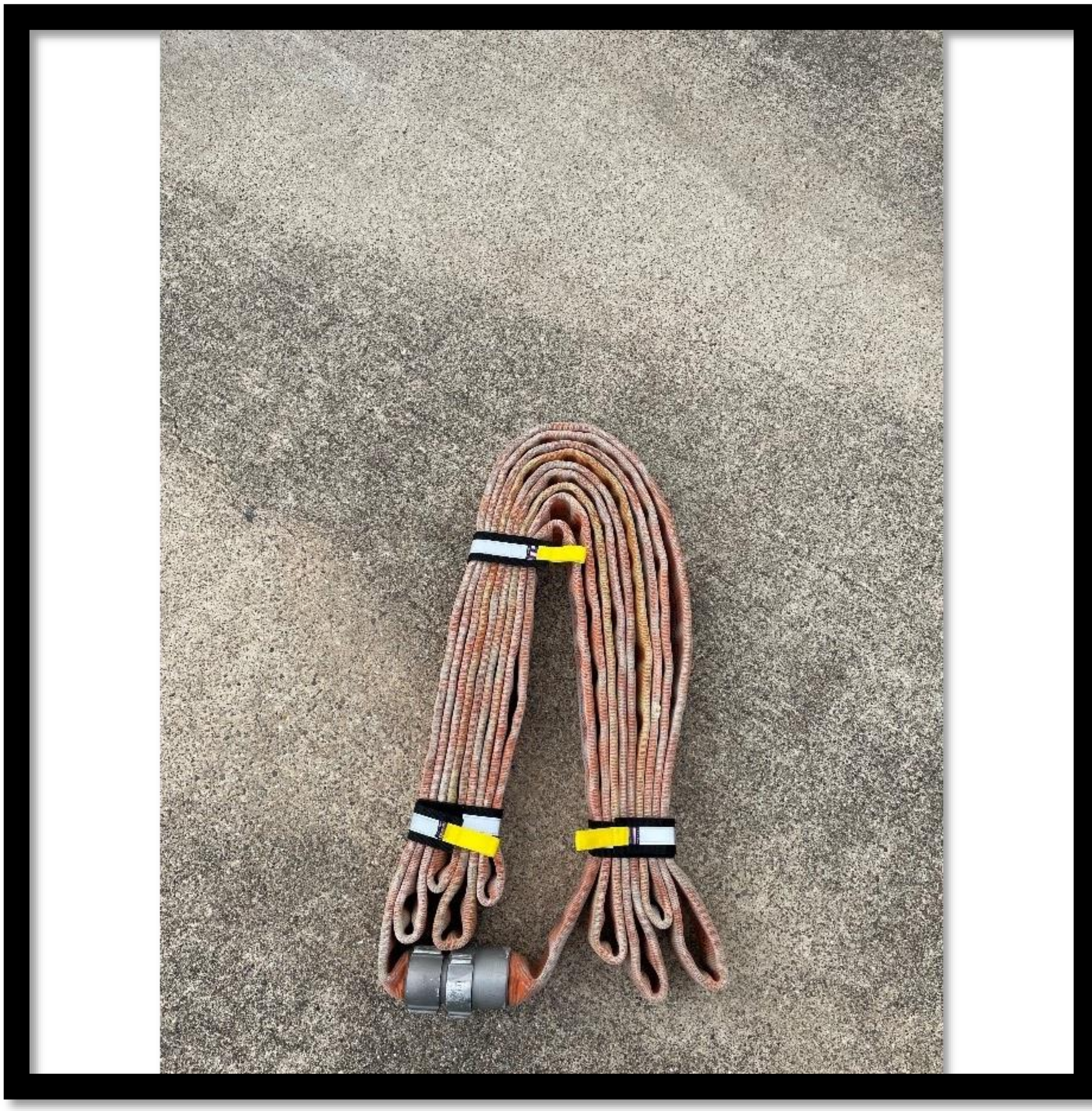


Figure 20

Attack Pack

When assembling the attack pack, you will follow the same steps that you did for the Supply Pack Steps 1-5. We will start with the sixth step for the attack pack.

Step 6: In contrast to the supply pack, we will not be connecting the couplings together, instead we will be connecting our 2.5” Chief XD Shutoff with 1 1/8” Tip to the male end of the hose and

Figure 21

placing it evenly with the edge of the load. (Figure 21.)

Step 7: When we connect this nozzle and place it evenly with the edge of the load, this may generate some excess hose on the top of the pack behind the nozzle. (Figure 22.) Instead of leaving a loose loop we will instead be bringing the slack back towards the female side of the load. (Figure 23.)

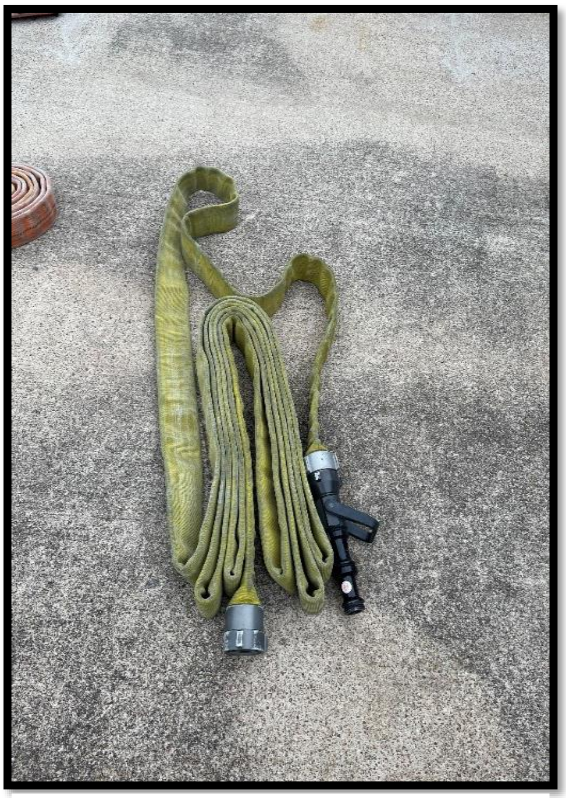


Figure 22

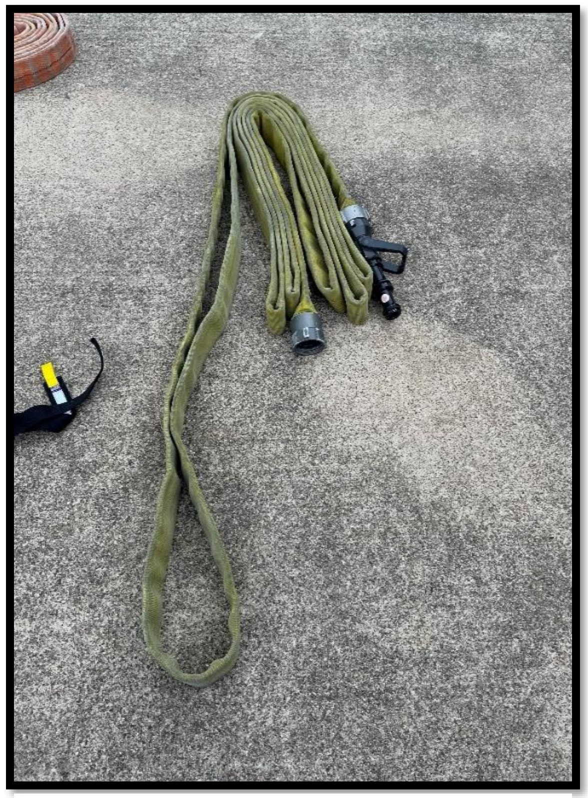


Figure 23

Figure 24

Step 8: Now that we have our slack pulled out, we will pull back the next fold on the female side of the load. (Figure 24.)

Figure 25

Step 9: We will then take the slack and fold it in-between that loop and the rest of the load. (Figure 25 & 26.)

Figure 26

Step 10: We will then place the load on its edge with the nozzle side up and place two straps through the innermost loop on the attack side of the pack. One strap will be directly on the nozzle shutoff and the other will be on the back side of the pack. (Figure 27 & 28.)

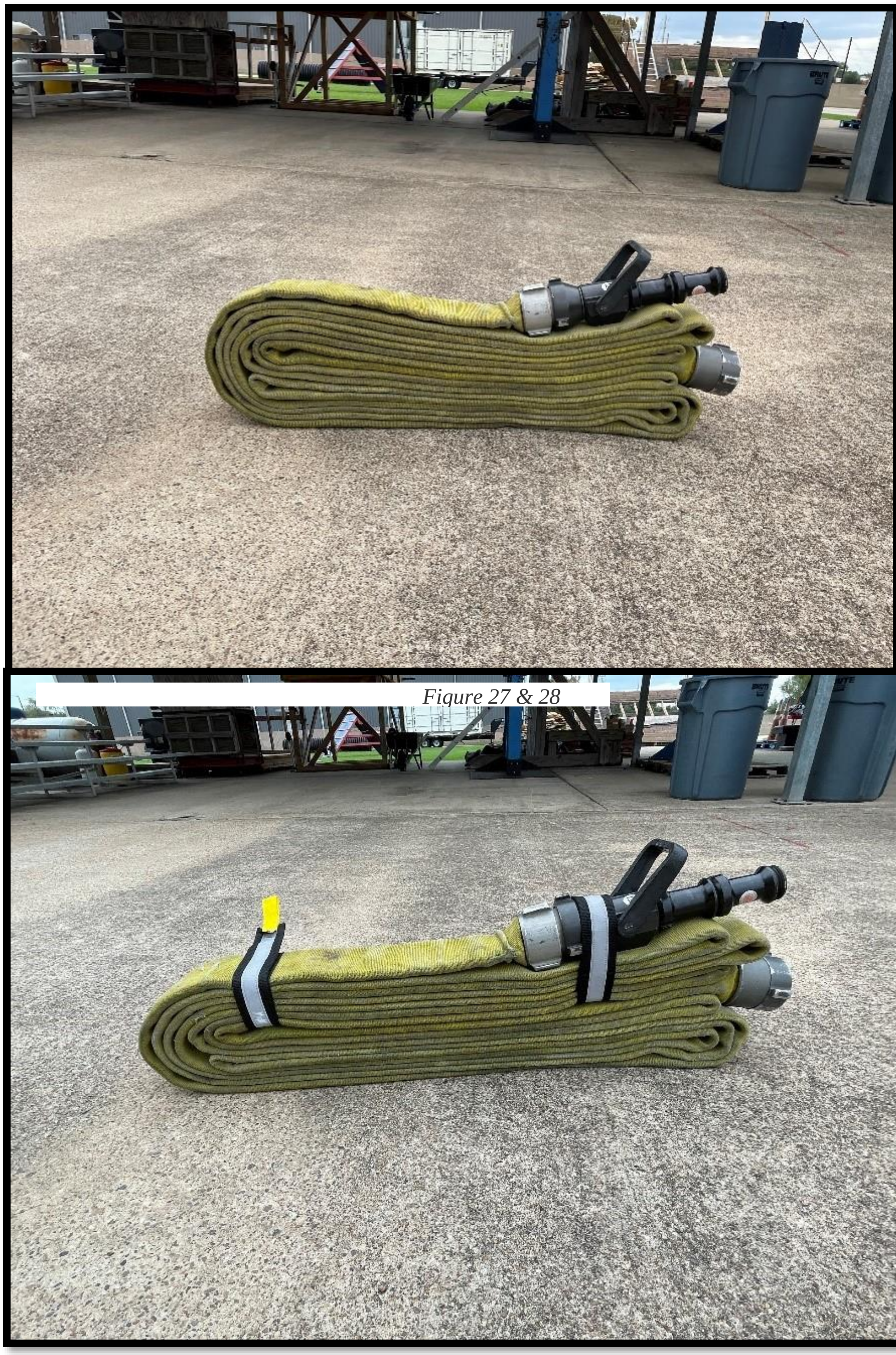


Figure 27 & 28

Step 11: The pack will then be placed on its other edge and a strap will be ran through the innermost loop on the supply side of the pack. (Figure 29 & 30.)

Figure 29

Figure 30

This completes the attack pack assembly, note that all straps are oriented on the same side of the pack and can all be removed without picking the pack up. (Figure 31.)

Figure 31

High Rise Hose Storage

The High-Rise Loads will be stored in one of two manners:

1. On the newest Engine Spec (Engine 2 & 5) the High-Rise Loads will be stored in the designated tray on the Officer Side of the Engine.

Figure 32

2. On all other Engines, the High-Rise Loads will be stored in the speedlay chutes.

High Rise Hose Deployment

High-Rise Packs

The Denver Load is the approved load for High-Rise and Standpipe Operations in the Pearland Fire Department. There are at least two different types of stretches that should be considered for every Standpipe operation: The Stairwell Stretch & Apartment Stretch. We will review the deployment steps for each stretch. After the attack stairwell and/or standpipe outlet is selected the crews should deploy the Denver load in the following manner:

Stairwell Stretch

Step 1: Once Firefighters reach the floor below the fire floor, they should drop their high-rise packs inside the public hallway or space on that floor (outside of the stairwell) this will be referred to as Drop Point #1. (Figure 33.)

Figure 33

Step 2: The Nozzle Firefighter should take the male coupling from the supply pack and attach it to the female coupling on the attack pack, as well as remove the straps from the supply pack while the Control Firefighter is hooking up to the Standpipe outlet. (Figure 34.)

Figure 34

Step 3: Once the Standpipe Outlet has been dressed the Control Firefighter will grab the female coupling of the supply pack and hook up to the Standpipe Outlet. (Figure 35 & 36.)

Figure 35

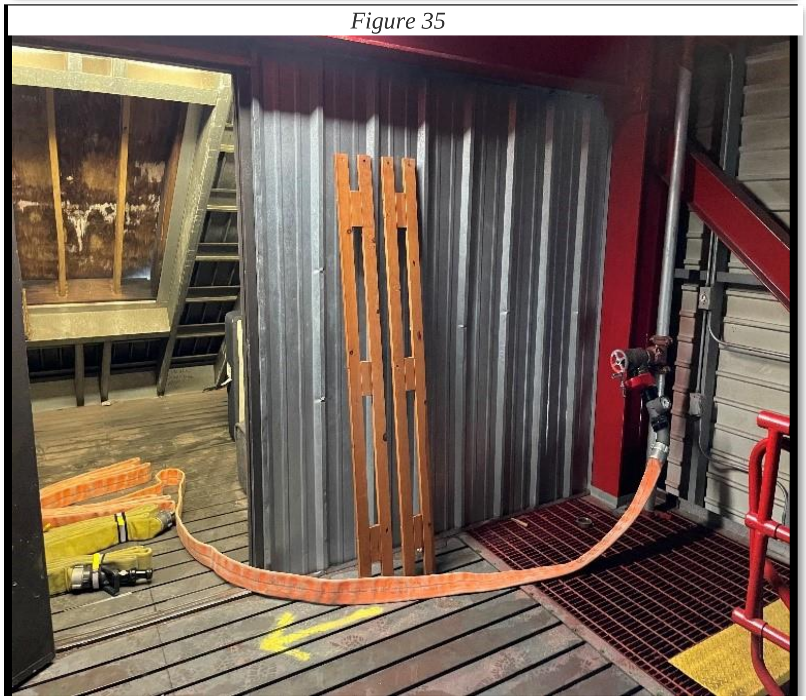


Figure 36

Step 5: The Nozzle Firefighter should proceed to the Fire Floor and drop the attack pack on the landing with the nozzle facing the direction of the fire door. Which we will refer to as Drop Point #2. They then should remove the straps from the bundle. (Figure 37 & 38.)

Figure 37

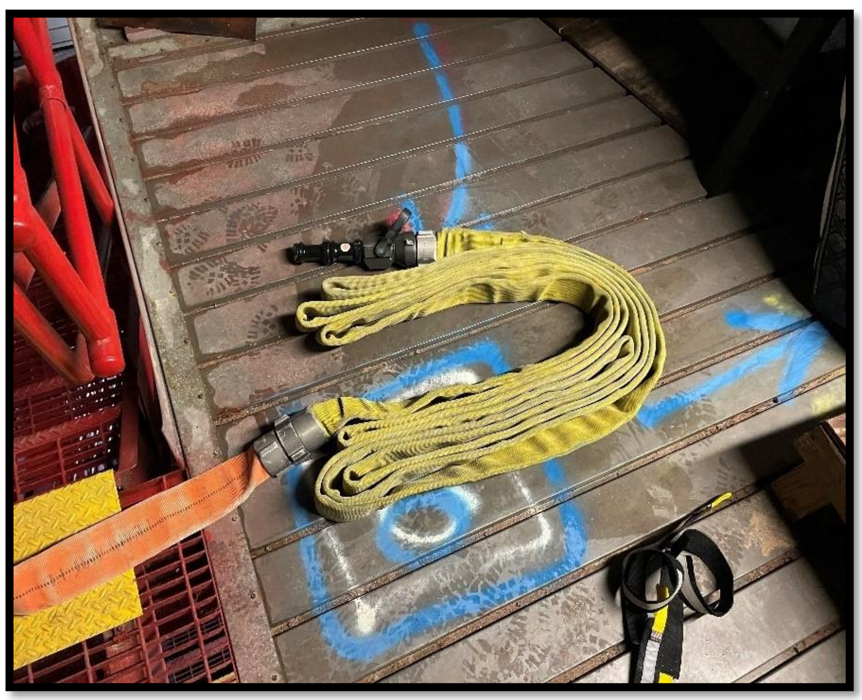


Figure 38

Step 6: The Nozzle Firefighter should grab the nozzle and approximately the top two loops of the bundle and advance the line up the next flight of stairs in front of the fire floor door. (Figure 39

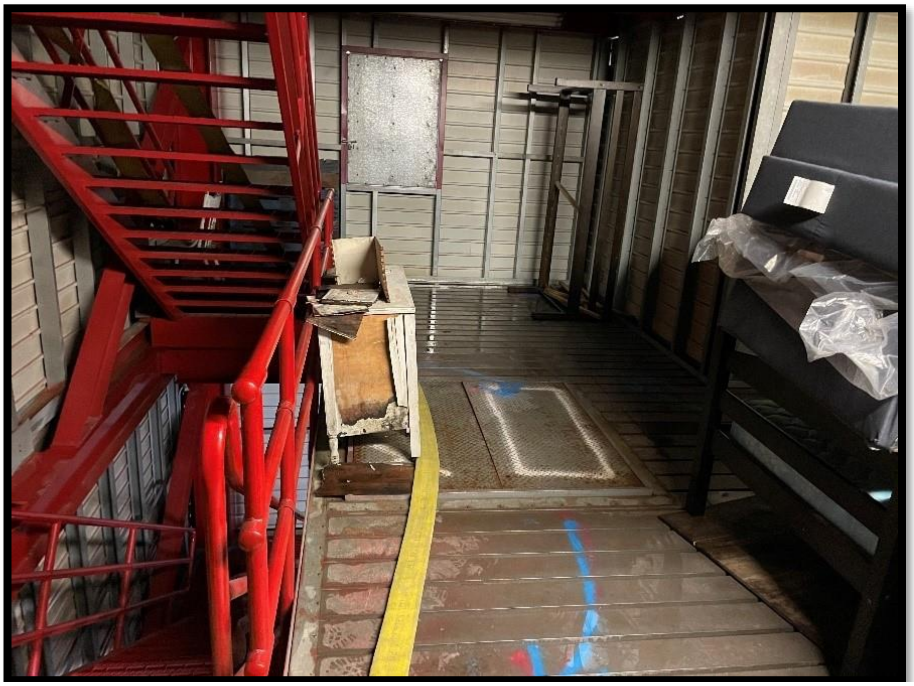


Figure 39

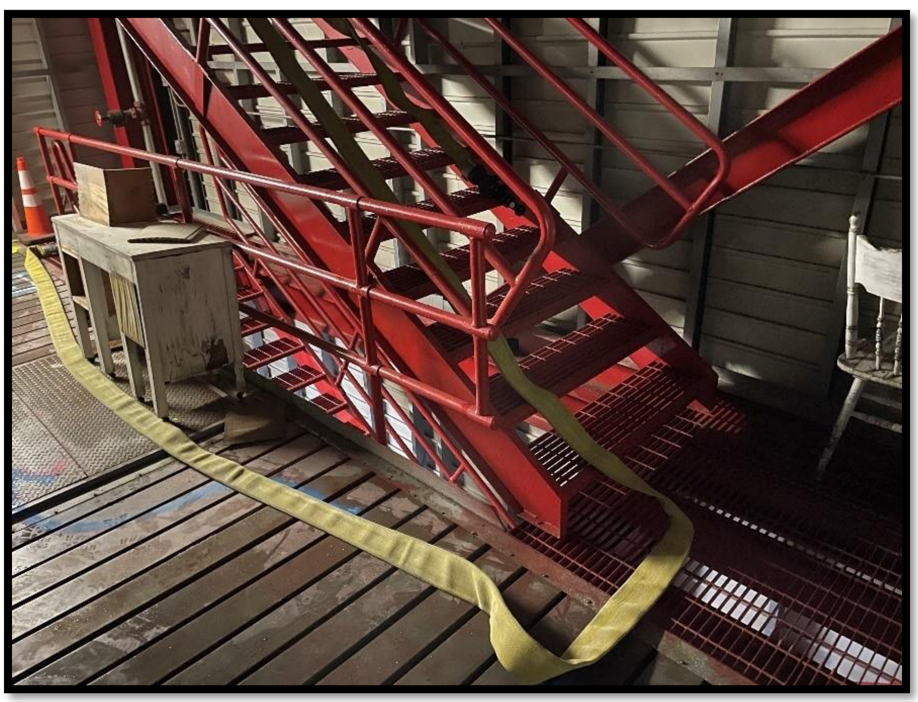


Figure 40

& 40.)

Step 7: Lastly, once the hose and his been flaked out up the flight of stairs the nozzle should be placed approximately 3-4 feet from the fire floor door. This is to prevent the Nozzle Firefighter from having their line pushed up against the fire floor door forcing them in a less defensible position. Once the Nozzle Firefighter is in place, another firefighter or the attack company Officer should kneel on the looped hose at the top of the stairs to prevent the hose from being pulled down when it's charged. Once everyone is in position the Nozzle Firefighter should call for water when ready.

Apartment Stretch

Step 1: Once Firefighters reach the floor below the fire floor, they should drop their high-rise packs inside the public hallway or space on that floor (outside of the stairwell) this will be referred to as Drop Point #1.

Step 2: The Nozzle Firefighter should take the male coupling from the supply pack and attach it to the female coupling on the attack pack, as well as remove the straps from the supply pack while the Control Firefighter is hooking up to the Standpipe outlet.

Step 3: Once the Standpipe Outlet has been dressed the Control Firefighter will grab the female coupling of the supply pack and hook up to the Standpipe Outlet.

Step 4: Simultaneously, the Nozzle Firefighter will shoulder the attack pack and grab approximately the top two loops of the supply pack to stretch out the remaining slack, this should be done in a controlled manner to ensure the hose is not being pulled away from the control man.

Step 5: The Nozzle Firefighter should proceed to the Fire Floor and drop the attack pack outside the fire compartment or apartment door on the hinge side of the door with the nozzle facing the direction of the fire door. Which we will refer to as Drop Point #2. They then should remove the straps from the bundle.

Step 6: The Nozzle Firefighter should deploy the bundle to where the hose is laid out attack over supply and can be easily advanced into the fire compartment.

Step 7: The Nozzle Firefighter should notify the Control Firefighter they are ready for water.

Step 8: Once the line has been charged, the Nozzle Firefighter should open the nozzle so that the Control Firefighter can set the pressure. The Nozzle Firefighter should continue to flow water until notified by the Control Firefighter the pressure is set.

High Rise & Standpipe Operations Riding, Tool, and Task Assignments.

PURPOSE: To set the operational priorities and responsibilities when responding, arriving, and operating at the scene of a High-Rise fire. PFD will consider a Highrise anything over 4 stories. Battalion should assume Lobby Sector on arrival.

<u>1st Due Engine</u>

- Identify attack stairwell.
- Fire Attack Officer, nozzle and control – GET WATER ON FIRE
- Take high-rise kit
- D/O- Water supply and hook into FDC for standpipe operations

<u>2nd Due Engine</u>

- Officer, Nozzle and Control link up with 1st due for fire attack.
- Take high-rise kits.
- D/O- Water Supply assist 1st due D/O

<u>3rd Due Engine</u>

- Officer, Nozzle and Control 2nd line/ backup line for fire attack
- Take high-rise kit
- D/O- Secure secondary water supply

<u>4th Due Engine- 2 Floors below fire</u>

- Officer to assume Staging sector until relieved or assigned
- RIT/On deck-2 floors below fire floor
- D/O with Crew
- Start tool cache
- Bottles to staging 2 floors below

<u>5thDue Engine</u>

- Supplement fire attack crews 2 floors below fire
- Bring additional attack line and high-rise kit
- D/O Stays with Crew

<u>1st Due Ladder</u>

- Low side to fire floor- Search
- High side to floor above -Search

<u>2nd Due Ladder</u>

- Low side fire floor. -Search
- High side floor above fire. -Search

3rd Due Ladder Chauffer with Truck

- Attack stairwell search- fire floor to roof
- Roof access for ventilation
- Top floor search-
- D/O-Set up for evacuation stairwell ventilation
- If fire is on low floor level, aerial ladder to fire floor

4th Due Ladder Chauffer with Truck

- Stage 2 floors below fire for manpower
- Take bottles and tools to staging
- Assist with Victim removal and secure stairwells on upper floors
- If fire is on low floor level, aerial ladder to fire floor

5th Due Ladder Chauffer with Crew

- Stage 2 floors below fire for manpower
- Take bottles and tools to staging

1st Due Medic

- Secure elevators and Control Room
- Assist with tool cache
- Extra SCBA bottles to tool cache
-

2nd Due Medic

- Evacuation stairwell starting at ground level working to fire floor.

3rd Due Medic

- Medical on staging floor
- Assigned to operate as needed by Staging sector

4th Due Medic

- EMS set up Triage area Rehab
- Assist with outside rescues

Squad 1

- Assist with accountability and Lobby Sector with BC
- EMS set up Triage area
- Rehab

Additional Units Arriving:

- Any additional arriving units will have their officer report to incident command for assignment
- Division supervisors will be assigned to the fire floor and staging as soon as possible
- Secondary staging will need to be identified for outside agencies to report to

Tool Assignments

Riding Assignment	Alarms	Fires	Rescues
Officer Radio ID Rig #	Radio – Big Light SCBA & TIC Forcible exit tool	Radio – Big Light SCBA & TIC Forcible exit tool	Radio
Nozzle Radio ID Rig# Nozzle	Radio – Big Light SCBA Can Pike pole (8’)	Radio – Big Light SCBA Attack Highrise Pack Pike Pole (8’)	Radio Can Pike pole (8’)
Control Radio ID Rig# Control	Radio – Big Light SCBA & TIC Irons	Radio – Big Light SCBA & TIC Supply High Rise Pack High Rise Kit Irons	Radio Irons
Backup (5th FF riding heavy) Radio ID Rig# Backup	Radio – Big Light SCBA Irons	Radio – Big Light SCBA & TIC Irons	Radio Irons
Driver Radio ID Rig# D/O	Radio Hook up to FDC	Radio Hook up to FDC	Radio Elevator Keys

Appendix A: Common Hydraulic Calculations

Necessary Formulas: Total Friction loss (TFL) = CQ^2L

- C = Coefficient of Friction
- Q = Fire flow in hundreds of GPM
- L = Length of hose

Pump Discharge Pressure (PDP) = TFL + Nozzle Pressure + Elevation loss (or gain) + Appliance Loss

Pre-determined pump pressures

200' Preconnect = **130 psi** Pump Discharge Pressure

Leader line pump pressures

- 100' 1 ¾" bundle = **90 psi** (40 psi friction loss, 50 psi nozzle pressure)
- Each section of 2 ½" used will be an additional **2.5 psi** when flowing a single 1 ¾"
- Each section of 2 ½" used will be an additional **10 psi** when 2 lines are flowing from a wyed line.
- Deck gun = **105 psi** pump discharge pressure

5" friction loss

- 2 psi per length at 500 gpm
- 8 psi per length at 1000 gpm
- 20 psi per length at 1,500 gpm
- 30 psi per length at 2,000 gpm

Appendix B: Engine Swap Procedures and Equipment

Reserve engine equipment

It is the intent that all reserve apparatus will be maintained in a “ready reserve state”. This would allow for the crew swapping into the reserve to only move over the equipment listed in this appendix. Crews will inspect the apparatus being put in service to ensure that it is equipped with the tools and equipment that are of the current standard. This may require moving over more equipment than desired, in order to maintain the current equipment standard.

When an apparatus is going out of service for more than 24 hours, all batteries and battery-operated equipment will be removed and either placed in service on the reserve apparatus or maintained at the firehouse to preserve battery life.

Move over equipment

- Electric PPV fan
- F500 Kit
- eDraulic combi tool
- High Rise Kit
- Tablet(s) w/ charger
- Gas meter w/ charger
- Airpacks x4
- EMS Equipment
- Radios w/ radio strap and lapel mic x4
- Ballistic Vests
- TIC x2
- Accountability passports
- Garage door opener
- RIT Pack

Appendix C: Engine Company Equipment

This section contains the make and model of all equipment that is standard for Engine Companies. It is the expectation that, when possible, this equipment will be present on all in service engines regardless of spec. Allotted quantities of each type of equipment is listed in section 1 of this manual. All exceptions to equipment types must be approved by the Assistant Chief of Operations.

- 1 ¾ and 2 ½ hose
  - Key Hose Tru-ID. 50’ lengths
- 5” Hose
  - Key ProFlow 100’, 50’, and 25’ lengths
- Thermal Imagers
  - Bullard TXS
- PPV Fan
  - Blowhard Quickee
- Combi-tool
  - Holmotro PCT 50
- Nozzles
  - 1 ½” Elkhart Chief XD Shutoff with 1 3/8” orifice
  - 7/8 solid stream Tip
  - Elkhart Chief XD Fixed gallonage 160 GPM fog tip
  - 2 ½” Elkhart Chief XD shutoff with integrated 1 3/16” tip
  - 1 1/8” solid stream tip with 1 ½” male threaded tip
  - Forestry Nozzle
- Ground Monitor
  - Akron Mercury Quick Attack
  - 1 3/8 solid stream tip
- Standpipe Kit
  - Elkhart standpipe kit with contents
- Vehicle stabilization struts
  - Holmotro V-Strut
- Battery powered tools and lights
  - Miketa 40V